

Review

Allylic substitution reactions with fluorinated nucleophiles

Artem A. Zemtsov, Vitalij V. Levin, Alexander D. Dilman*

N. D. Zelinsky Institute of Organic Chemistry, 119991 Moscow, Leninsky prosp. 47, Russian Federation



ARTICLE INFO

Article history:

Received 6 December 2021

Accepted 1 February 2022

Available online 16 February 2022

Keywords:

Organofluorine

Allylic substitution

Transition metals

Fluorinated nucleophiles

ABSTRACT

The importance of organofluorine compounds has spurred the development of methods aimed at their synthesis. In this review, carbon–carbon bond forming reactions between allylic electrophiles and fluorinated nucleophiles are discussed. Compared to classical saturated alkyl halides, allylic electrophiles have increased reactivity especially in transition metal mediated processes, which is associated with propensity of the allylic systems to interact with low-valent transition metal complexes. Due to specific electronic effects, fluorine atoms may have a profound impact on the stability and reactivity of carbanionic reagents. In the review, nucleophilic reagents bearing one, two or three fluorine atoms at the nucleophilic carbon are considered. Various types of fluorinated reagents including silicon, copper, and zinc derivatives, employed both in catalytic or stoichiometric variants, as well as reagents having a stabilized fluorinated carbanion, are presented. The review is comprehensive, and covers the literature starting from the earlier work on conventional nucleophilic substitution reactions up to recent transition metal catalyzed processes.

© 2022 Elsevier B.V. All rights reserved.

Contents

1. Introduction	2
2. Reactions of monofluorinated carbanions	2
2.1. Reactions of anions stabilized by electron-withdrawing groups	2
2.1.1. Derivatives of malonic esters and ketoesters	2
2.1.2. Derivatives of esters, amides, ketones and azomethines	3
2.1.3. Anions stabilized with phosphorus	5
2.1.4. Miscellaneous stabilized carbanions	6
2.2. Reactions catalyzed by transition metal complexes	7
3. Reactions of difluorinated carbanions	9
3.1. Reactions of carbanions stabilized by electron withdrawing groups	9
3.1.1. Organolithium compounds	9
3.1.2. Activated silicon complexes	9
3.1.3. Claisen rearrangement	10
3.2. Reactions catalyzed by transition metals	10
3.2.1. Cadmium reagents	10
3.2.2. Copper reagents	11
3.2.3. Palladium catalysis	14
4. Trifluoromethylation and perfluoroalkylation reactions	15
4.1. Reactions of perfluorinated magnesium, cadmium and silver derivatives	16
4.2. Reactions of perfluoroalkylsilicon derivatives	16
4.3. Reactions catalyzed by palladium	17
4.4. Reactions of copper (I) derivatives	18
4.4.1. Reductive metalation	18
4.4.2. Transmetalation of zinc and cadmium reagents	18

* Corresponding author.

E-mail address: adil25@mail.ru (A.D. Dilman).

4.4.3.	Transmetalation of activated silicon complexes.....	19
4.4.4.	Synthesis from the sources of difluorocarbene.....	20
4.4.5.	Isolated copper complexes.....	21
5.	Conclusion.....	22
	Declaration of Competing Interest.....	22
	References.....	22

1. Introduction

A single fluorine atom or a fluorinated group can dramatically change physical, chemical and biological properties of organic molecules [1,2]. For this reason, organofluorine compounds have received great attention in a number of applied disciplines such as medicinal chemistry [3–5], agrochemistry [6,7], and materials sciences [8,9]. Among many approaches for the direct introduction of fluorinated fragments [10,11] processes of nucleophilic fluoroalkylation have become particularly widespread which is associated with ready availability of fluoroinated carbanionic reagents [12–17]. Concerning electrophilic component, allylic and propargylic substrates demonstrate excellent reactivity in coupling with fluorinated reagents via both transition metal mediated and classical nucleophilic substitution mechanisms. Herein we present a comprehensive overview of fluoroalkylation processes involving these classes of electrophiles.

A general feature of allylic electrophiles is an opportunity to undergo substitution at α - and γ -positions. The regioselectivity of the reaction can be affected by many factors such as stereoelectronic effects in the substrate molecule, as well as the particular reaction conditions. The reaction outcome can also be affected by the nature of nucleophilic component. It should also be noted that the stability and the reactivity of fluoroalkyl anions can significantly differ from their non-fluorinated analogs. Indeed, fluorine atoms themselves have an impact on adjacent anionic center due to the high electronegativity and electrostatic repulsion of electron pairs [18,19]. For this reason, the reactions in this review are divided into sections in accordance with the number of fluorine atoms connected to the nucleophilic center.

2. Reactions of monofluorinated carbanions

Nucleophilic reactions of stabilized α -monofluorinated carbanions constitute a reliable approach for the introduction of a fluorine-containing fragment. The presence of one or two electron-withdrawing groups sufficiently reduce the tendency of the anionic center to eliminate fluoride [20], which makes it possible to use such compounds for direct fluoroalkylation without transition metals. At the same time, palladium or iridium catalysis may also be used to control the stereoselectivity of the process or to involve new types of allylic substrates.

Fluorine may have notable and in some cases even counterintuitive effect on the reactivity of carbanions. For example, introduction of a fluorine atom at the carbanionic center is expected to increase the stability of the carbanion. At the same time, there have been reports suggesting that α -fluorine substitution enhances the nucleophilic reactivity of carbanions [21,22]. Quantitative kinetic measurements of Mayr's nucleophilicity parameters of carbanions stabilized by two electron withdrawing groups (ester or sulfonyl) showed that mono-fluorine substituted carbanions are *more* reactive compared to non-fluorinated counterparts [20] (Scheme 1). The difference in rate constant may achieve three orders of magnitude, as evidenced for cyclic bis(sulfonyl)-stabilized carbanions (bottom structures). It is likely that effect of the fluorine on pyramidalization of the carbanionic center plays important role in

these observations, though influence of the nature of electron withdrawing group should also be taken into consideration [23–25].

Non-stabilized monofluorinated lithium or magnesium reagents are very labile and cannot be prepared as in a stable form even at low temperature. The nature of the metal, as well as of coordinating solvent, may have significant influence of the reagent stability [26]. Fluoromethylolithium was successfully trapped if generated in the presence of aldehydes or ketones or other π -electrophiles [27,28], but reactions with allylic electrophiles have not been described.

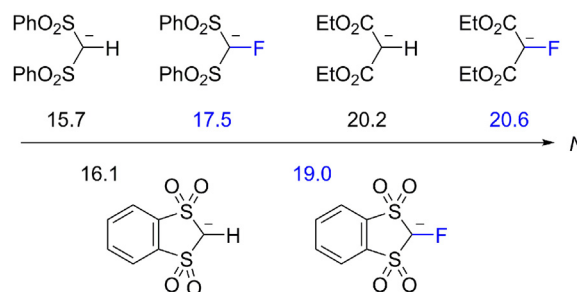
2.1. Reactions of anions stabilized by electron-withdrawing groups

2.1.1. Derivatives of malonic esters and ketoesters

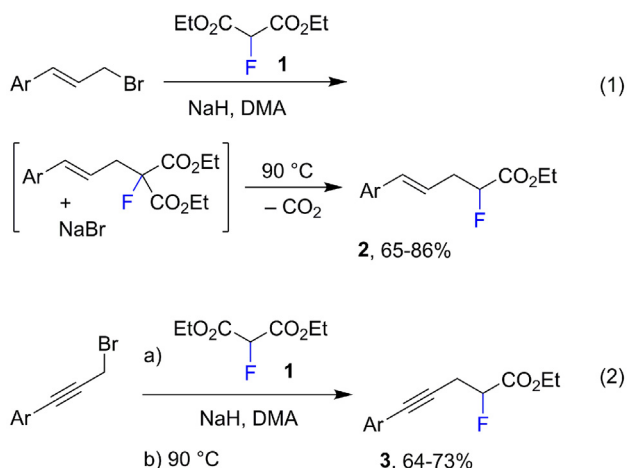
Diethyl fluoromalonate **1** is less reactive in alkylation processes than typical non-fluorinated malonic esters (Scheme 2). Allylation of malonate **1** in the presence of sodium ethylate proceeds in low yield and is accompanied by many side processes. However, with a stronger base such as sodium hydride in DMF, the reaction of **1** with allyl bromide was carried out with 79% yield [29,30]. Sodium bromide, which forms as a by-product can also serve as a promoter of further decarboxylation process. Such tandem fluoroalkylation/Krapcho reaction of allyl and propargyl bromides led to the esters **2** and **3** with good to high yields [31].

An asymmetric variant of this transformation was firstly demonstrated on a monoether **4** bearing 8-phenylmenthyl auxiliary group (Scheme 3). Analogs of compound **4** bearing two ester groups were ineffective in this transformation due to the low activity of the enolate anion. At the same time, the half-ester treated with an excess of lithium hexamethyldisilazide (LiHMDS) in tetrahydrofuran forms more nucleophilic dianion, which readily interacts with allyl bromide. Other bases were also evaluated, but good diastereoselectivity was obtained only under lithiation conditions, which is attributed to the stability of one of the conformations of lithium enolate (shown in Scheme 2). Subsequent treatment of the allylation product with diazomethane led to olefin **5** in 93% yield and high diastereoselectivity [32].

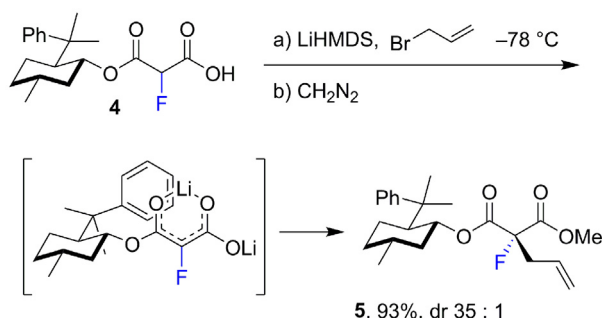
In a more recent study, a catalytic version of an asymmetric allylation of α -fluoromalonates was presented. Under phase-transfer conditions the reaction of diphenylmethyl *tert*-butyl fluoromalonate **6** with allyl bromide gave product **7** in 82% yield and



Scheme 1. Mayr's nucleophilicity parameters.



Scheme 2. Reactions of diethyl fluoromalonate.



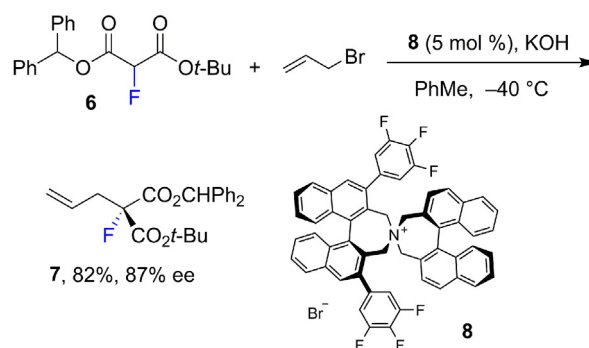
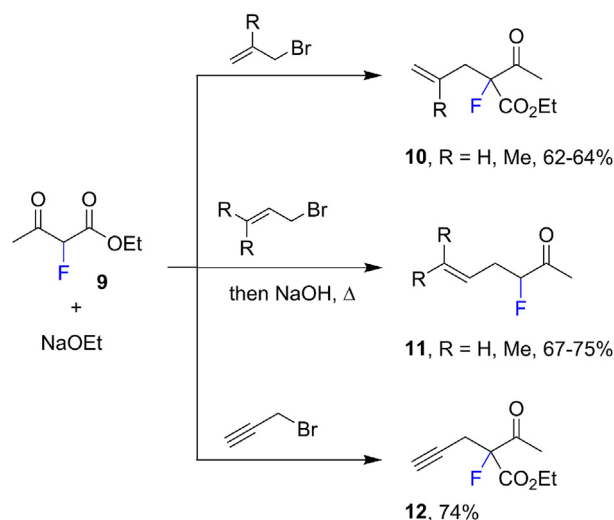
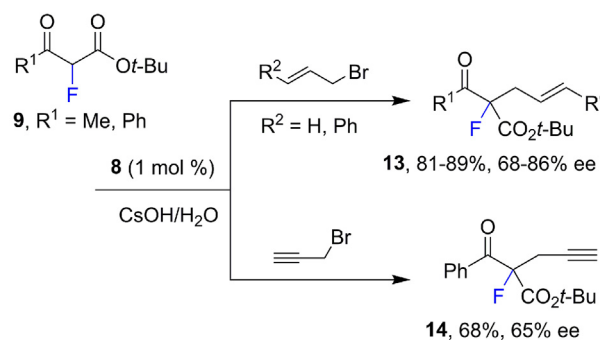
Scheme 3. Allylation of 8-phenylmenthyl fluoromalonate.

87% enantiomeric excess (Scheme 4). Quaternary ammonium salt **8** (Maruoka catalyst) was used as a chiral phase-transfer catalyst. The authors demonstrated that the use of dry potassium hydroxide instead of its 50% aqueous solution does not affect the yield of the process, but significantly reduces the reaction time [33].

Fluoroacetoacetic ester **9** can react with primary allyl and propargyl bromides in the presence of sodium ethylate with the formation of products **10–12** (Scheme 5). The reaction rate greatly depends on the steric accessibility of the electrophilic center and cannot be increased by higher temperatures due to the deacetylation of compound **9**. At the same time, in the absence of steric hindrances S_N2 -substitution products **10–12** were obtained in good yields [34].

There is also an asymmetric version of the reaction of α -fluoroketoesters with primary allyl and propargyl bromides in the presence of phase-transfer Maruoka catalyst **8** (Scheme 6). The fluorinated derivative of benzoylacetic ester **9** reacted with allyl halides leading to products **13** in high yields and enantioselectivities; slightly worse results were observed for a derivative of acetoacetic ester. A noticeable deterioration in yield and enantioselectivity was observed in the propargylation reaction affording alkyne **14** [35].

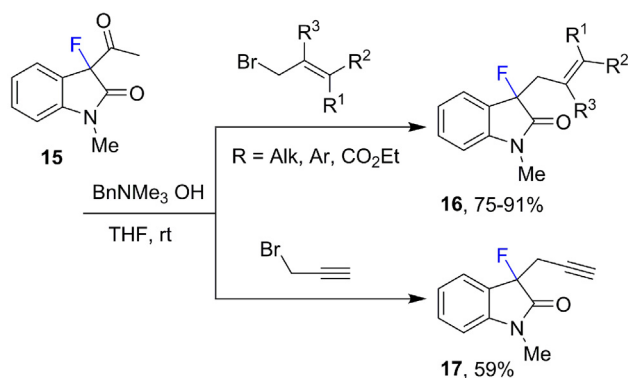
3-Acetyl-3-fluoro-2-oxindole **15** easily undergoes deacetylation under basic conditions and then can interact with various C-electrophiles (Scheme 7). In the reactions with functionalized alkyl halides α -substitution products **16** were selectively obtained in high yields. In the case of cyclohexenyl bromide, the corresponding reaction product was formed as a 1:1 mixture of diastereoisomers. The reaction with propargyl bromide led to the α -substitution product **17** in 59% yield [36]. Starting oxindoles **15** can be effectively

Scheme 4. Phase-transfer catalytic allylation of α -fluoromalonate.Scheme 5. Allylation and propargylation of ethyl α -fluoroacetoacetate.Scheme 6. Asymmetric reactions of α -fluoroacetoacetates.

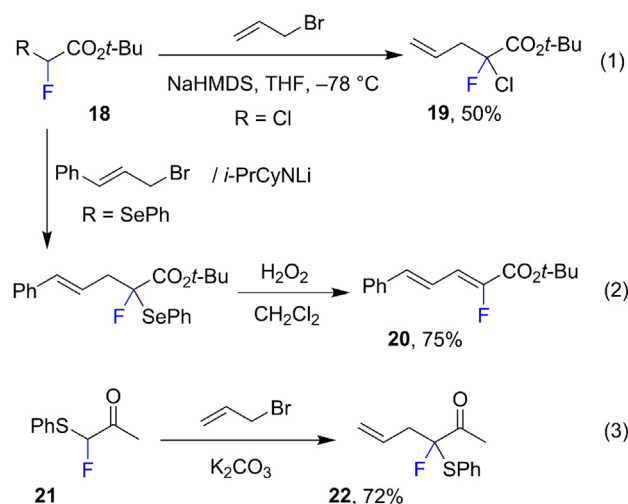
obtained by electrolytic dehydrogenative coupling of corresponding *N*-acylated anilines [37].

2.1.2. Derivatives of esters, amides, ketones and azomethines

The first example of allylation of α -fluoro-substituted esters was described in the investigation of the reactions of different dihaloacetates with C-electrophiles. *tert*-Butyl chlorofluoroacetate (**18**, R = Cl) reacted with allyl bromide upon deprotonation with sodium hexamethyldisilazide at low temperature (Scheme 8, equation 1). The yield of the product **19** was 50%, which was significantly lower than the yields of similar dichloro and dibromo



Scheme 7. Deacilative allylation and propargylation reactions.

Scheme 8. Allylation of α -fluorinated carbonyl compounds.

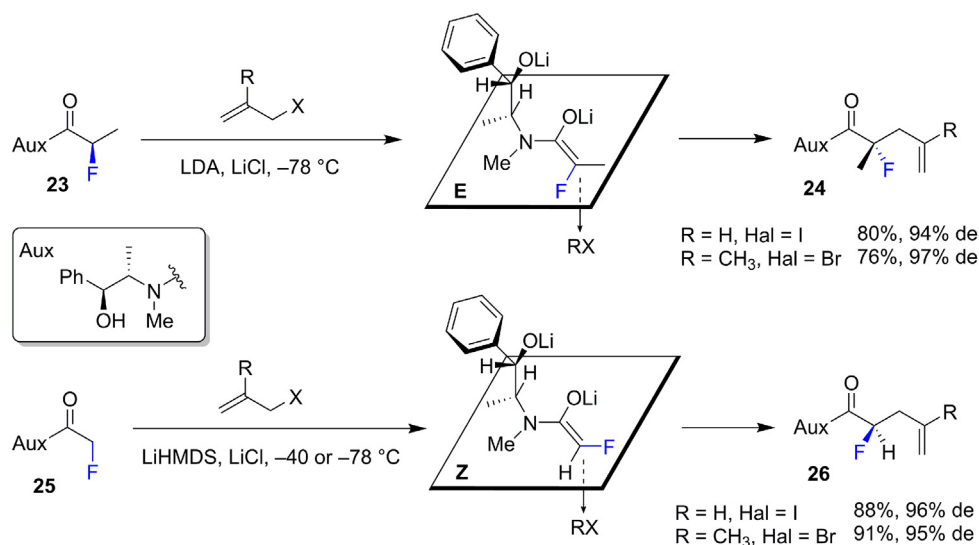
derivatives. The authors explained this fact by destabilizing effect of lone electron pairs of fluorine atom on the enolate anion [38].

The reaction of (phenylselenenyl)fluoroacetate (**18**, R = SePh) with cinnamyl bromide upon deprotonation with lithium dialkylamide

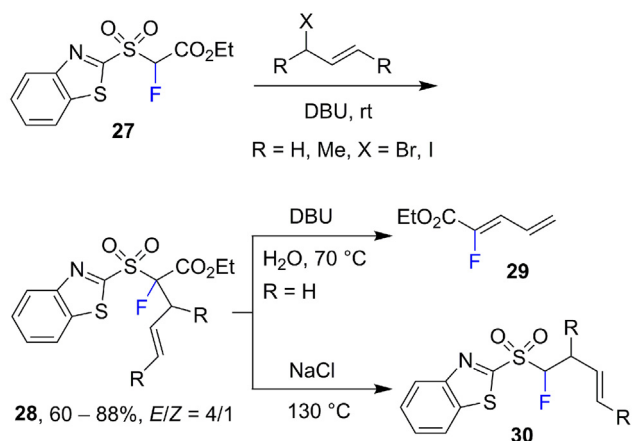
was successfully implemented as an alternative to direct electrophilic fluorination (Scheme 8, equation 2). The obtained geminal fluoroselenide **19'** was converted into diene **20** under oxidative conditions in 75% yield for two steps [39]. Also in the reaction of ketone **21** bearing the phenylsulfenyl group with allyl bromide, the product **22** was obtained in 72% yield (Scheme 8, equation 3) [40].

Allylation of a fluorinated propionamide of pseudoephedrine **23** in the presence of lithium diisopropylamide and an excess of lithium chloride resulted in products **24** in high yields and excellent diastereoselectivities (Scheme 9, equation 1). Less substituted acetamide **25** upon treatment with lithium hexamethyldisilazide and subsequent reaction with allyl or prenyl halides was also converted into compounds **26** with perfect selectivities (Scheme 9, equation 2). The authors noted that such stereochemical results in both cases may indicate the formation of stable forms of enolates **E** and **Z** at the stage of lithiation [41,42]. This is probably the reason why the choice of the base has a crucial effect on the selectivity of the reaction. Thus, replacement of LDA by LiHMDS in the case of propionamide **23** led to the decrease of diastereomeric excess to 10–24%. Derivative of fluoracetic acid **25** also demonstrated certain sensitivity to the nature of the base, showing the decrease of diastereomeric excess to 64%. As a possible explanation, the authors referenced to their earlier studies, according to which the *cis*-form is energetically preferred for unsubstituted halogen enolates. The formation of the *trans*-form is determined by kinetic control, which leads to the strong dependence of the selectivity on the strength and steric hindrance of the base.

Allylation of benzothiazolylfluorosulfonyl acetate **27** was proposed as one of the methods for the synthesis of sulfones **28**, promising substrates for the Julia–Kocienski reaction (Scheme 10) [43]. The reaction of **27** with allyl and methallyl halides was carried out at room temperature in the presence of DBU as a base; as a result, olefins **31** were obtained in high yields. The reactions of 1-bromo-2-butene and 3-bromo-1-butene with reagent **27** led to the same regioisomer of **28**, which indicates the prevailing effect of the steric factor on the direction of substitution. Also in both cases an identical *E/Z*-selectivity was observed with a significant predominance of the *E*-isomer [44]. After standard basic hydrolysis of the ether **28** elimination of the sulfonate group happened instead of decarboxylation and the diene **29** was formed. At the same time, decarboxylated sulfones **30** were successfully obtained under the conditions of the Krapcho reaction.



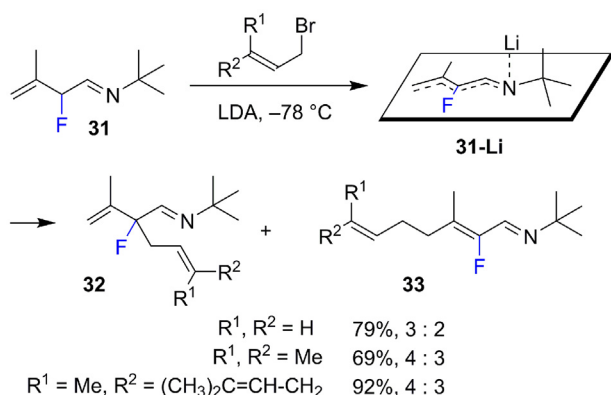
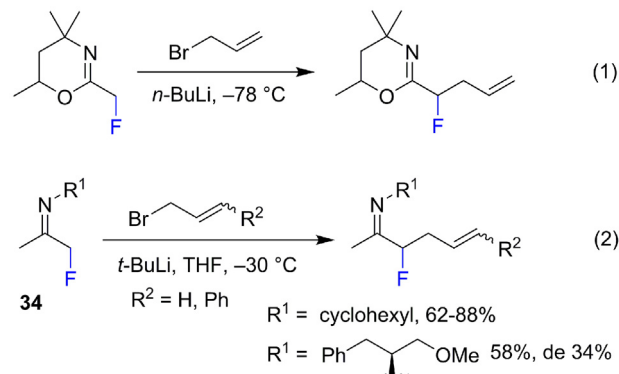
Scheme 9. Allylation of pseudoephedrine derivatives.



Scheme 10. Alkylation of benzothiazolylfluorosulfonyl acetate.

Imine **31** derived from β,γ -unsaturated 2-fluoro-3-methylbut-3-enal may be easily deprotonated with lithium diisopropylamide resulting in a stable 3-fluoro-1-azapentadienyl lithium salt **31-Li**. This salt has significant conformational flexibility and can exist in *U*-, *S*- and *W*-shaped forms (Scheme 11 depicts the most stable *W*-form). Different electrophiles may interact with lithium 1-azapentadienyl intermediate **31-Li** at one of three positions: at the nitrogen atom, at the fluorinated α -carbon atom or at the terminal γ -carbon atom. The reactions of lithiated azomethine **31** with allyl, prenyl, and geranyl bromide were studied [45]. In these processes, the mixture of isomeric alkenes **32** and **33** was formed in good total yield, which indicates the competition between the attack on α - and γ -positions. The ratio of the products was approximately equal, with a slight predominance of the branched isomer **32**. Linear alkenes **33** were stereochemically homogeneous – the resulting multiple bond in all cases had *Z*-configuration, which confirms the preference of *W*-conformation of **31-Li**. The authors also noted that in a similar reaction of prenyl bromide with 3-methyl-1,3-butadiene-1-imine, the non-fluorinated analogue of **31**, an exclusively branched isomer was formed.

Alkylation of 2-fluoromethyl-4,4,6-trimethyl-1,3-oxazine with allyl bromide upon deprotonation with butyllithium was described (Scheme 12, equation 1). The reaction proceeded in 66% yield [46]. Imines derived from ketones can also react with allyl electrophiles; however, in this case regioselectivity problems arise. It was shown that *N*-cyclohexylimine of fluoroacetone **34** in reaction with allyl halides upon deprotonation with *tert*-butyllithium at -30 °C mainly formed products resulting from the substitution at the flu-

Scheme 11. Alkylation of imine **31**.

Scheme 12. Selective allylation of fluoromethyl azomethines.

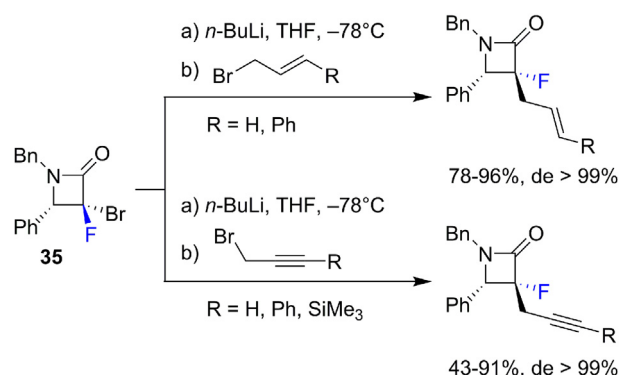
orinated carbon atom (Scheme 12, equation 2). When LDA was used as a base at -78 °C, the opposite regioisomer was formed. The process was also carried out in a stereoselective version with the imine derived from fluoroacetone and (*S*)-phenylalaninol *O*-methyl ether [47].

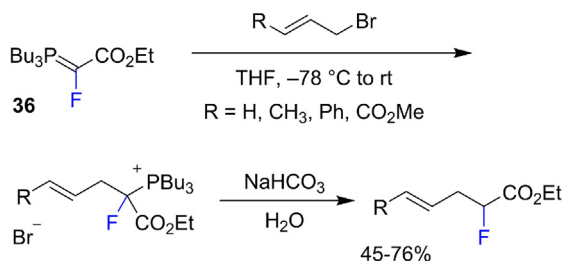
It was shown that α -bromo- α -fluoro- β -lactams **35** may react with various allyl and propargyl bromides under lithiation conditions. In this processes, α -substitution products were obtained in good yields as a single diastereomer (Scheme 13). High selectivity was explained by steric factors, since interaction of electrophile with the flat lithium enolate always proceeds in the *trans*-position relative to the sterically more loaded phenyl substituent [48]. A similar process of allylation of non-bromosubstituted lactams upon deprotonation with LDA was also described [49].

2.1.3. Anions stabilized with phosphorus

Fluorophosphorane **36** may be easily generated by interaction of ethyl bromofluoroacetate and tributylphosphine, followed by deprotonation of the resulting quaternary phosphonium salt [50]. Despite the low activity of phosphorane **36** in conventional alkylation reactions, its interaction with allyl bromide and its 1-alkyl-substituted derivatives proceeds without any difficulties (Scheme 14). In this case, phosphonium salts were obtained and further hydrolyzed under basic conditions to give esters. The yields of products in a series of reactions with various allylating reagents were in a range of 45–76% based on ethyl bromofluoroacetate, a precursor of phosphorane **36**. The substitution proceeded regio- and stereospecifically maintaining the original *E*-configuration of the double bond of alkenes [51].

One of the methods for the synthesis of derivatized α -fluoroalkylphosphonates is based on alkylation of corresponding carban-

Scheme 13. Reactions of lithiated β -lactams.



Scheme 14. Alkylation of fluorophosphorane.

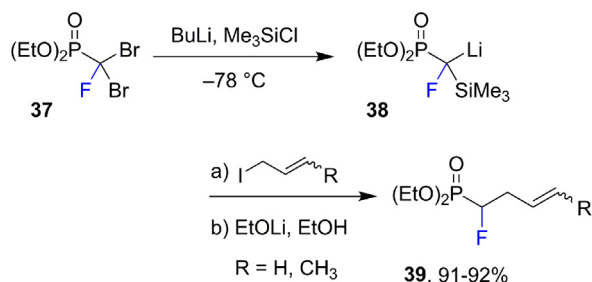
ions. The main problem of this approach is the tendency of α -fluorophosphonoalkyl anions to form carbenoid species by α -elimination of phosphite or fluoride [52]. In literature, there are only a few examples of reactions of lithium [53] or zinc [54] monofluoromethylphosphonates with allyl bromide with the yields of 46% and 80%, respectively. At the same time, the vast majority of alkylation methods assume the presence of additional stabilizing or protecting groups.

Tandem lithiation/allylation of (dibromofluoromethyl)phosphonate **37** may be done in one step in high yield (Scheme 15). When phosphonate **37** was treated with two equivalents of butyllithium in the presence of one equivalent of chlorotrimethylsilane, silylated anion **38** is formed as a result of bromine-lithium exchange and silylation steps. The reaction of **38** with allyl or crotyl iodide followed by protodesilylation led to products **39** in excellent yields [55,56]. It was also noted that the reagent **38** showed higher nucleophilic activity in comparison with a non-fluorinated analogue.

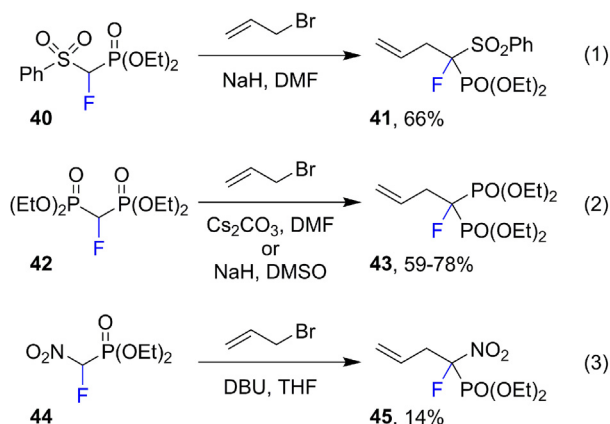
α -Fluorophosphonates bearing an additional electron withdrawing substituent demonstrate remarkably high stability upon deprotonation. Thus, sulfonylphosphonate **40** may be considered as one of the first examples of fluoromethylene synthons (Scheme 16, equation 1). Reagent **40** readily reacted with allyl bromide in the presence of sodium hydride resulting in the formation of the bifunctional product **41** in 66% yield. Sulfonyl or phosphonate groups may be selectively removed in further transformations [57].

Bisphosphonate **42** was reacted with allyl bromide in the presence of cesium carbonate in dimethylformamide (or sodium hydride in dimethyl sulfoxide) at room temperature (Scheme 16, equation 2). The highest yield of product **43** was achieved with cesium carbonate; however, the reaction time was about six days. Attempts to accelerate the reaction by heating it to 50 °C led to the drop of the yield to 14%, which was explained by the decomposition of cesium salt of phosphonate **42**. The reaction in the presence of sodium hydride proceeded within one hour with a moderate yield [52].

Another example of α -fluorophosphonates with a relatively stable carbanion is nitro compound **44** (Scheme 15, equation 3). The maximum activity of **44** in alkylation reactions was achieved



Scheme 15. Silylation/allylation of (dibromofluoromethyl)phosphonate.

Scheme 16. Alkylation of substituted α -fluorophosphonates.

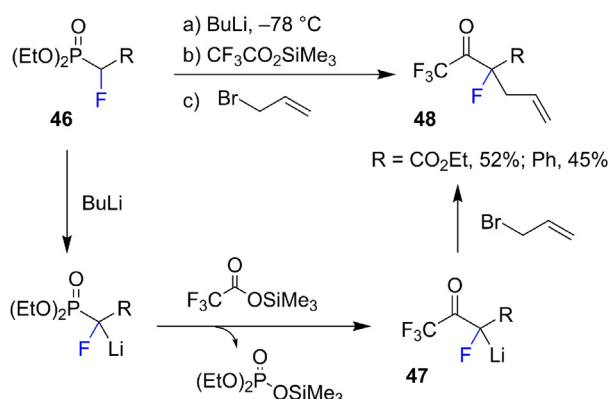
with DBU in tetrahydrofuran; however, the yield of product **45** in reaction with allyl bromide did not exceed 14%. Raising the temperature above 0 °C was also ineffective and was accompanied by the decomposition of compound **44** [58].

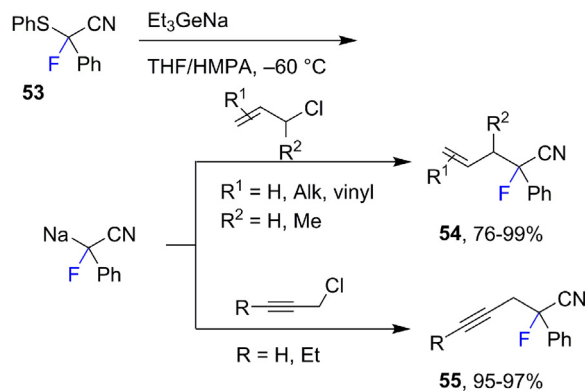
2.1.4. Miscellaneous stabilized carbanions

Fluorophosphonates **46** served as key compounds in an unconventional method for the synthesis of tetrafluoroketones **48** (Scheme 17) [59]. Benzyl and ethoxycarbonyl fluorophosphonates **46** reacted with trimethylsilyl trifluoroacetate upon deprotonation with butyllithium. As a result, phosphonate and trifluoroacetyl residues exchanged with the formation of enolate **47**. Further boiling of the reaction mixture in the presence of allyl bromide led to the formation of ketones **48** in moderate yields [60].

2-Fluoro-2-phenyl-2-phenylthioacetone **53** reacts with allyl chlorides and propargyl bromides upon treatment with triethylgermyl sodium affording α -substituted products **54** and **55** (Scheme 18). The germyl sodium reagent in the presence of HMPA at -60 °C activates the C-S bond of nitrile **53** with the formation of a reactive intermediate, which then acts as an equivalent of the cyanofluorophenylmethyl anion. The interaction of this intermediate with primary allyl ($R^2 = H$) and propargyl chlorides proceeded under mild conditions in nearly quantitative yields and complete retention of the configuration of the double bond. In the case of secondary 3-chlorobut-1-ene, the main α -substitution product **54** was formed in 76% yield along with its γ -substituted regioisomer, which was isolated in 24% yield [61].

The choice of the activator plays a pivotal role in this procedure, which was demonstrated on the reaction of nitrile **53** with cin-

Scheme 17. Dephosphorylation/allylation of α -fluorophosphonate.

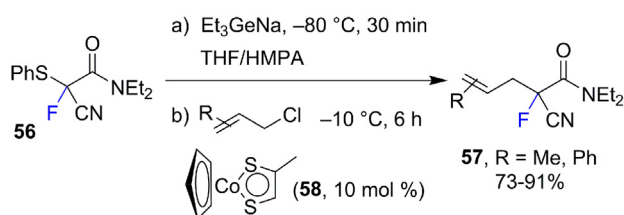


Scheme 18. Reactions of 2-fluoro-2-phenyl-2-phenylthioacetonitrile.

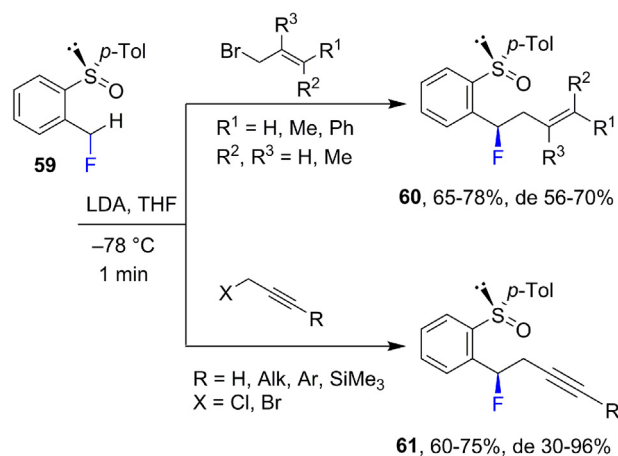
namyl chloride. When the process was promoted with the germyl lithium or sodium reagents, the yield of the target product decreased to 35% and 6% respectively, while maintaining the complete conversion of the starting cyanide. Variation of different bases gave positive results only in cases of butyllithium (36% yield) and lithium naphthalenide (also 36% yield), which is known as a strong single-electron reducing agent. The authors explained the exceptional ability of the germyl sodium to activate the C–S bond by a high affinity of germyl anion to the sulfur atom. The counter ion was proposed to affect the stability of the active intermediate; however, the exact structure of the latter was not established.

Phenylthio-substituted amide **56** with a structure similar to that of **53** may also be activated with triethylgermyl sodium; however, the resulting sodium salt does not undergo allylation due to its higher stability (Scheme 19). Various complexes of cobalt, nickel, palladium, platinum, rhodium and ruthenium were tested as the catalysts in this process. The best result was achieved with a cobalt dithiolene complex **58**. It was demonstrated that the substituents both in the thiol fragment and in the cyclopentadienyl ring have a significant effect on the catalyst efficiency. As a result, in the reaction of **56** with allyl chlorides in the presence of 10% of the catalyst, α -substitution products **57** were obtained in high yields [62].

It is known that sulfoxide group can affect regio- and stereoselectivity of the substitution at the fluorinated carbanion center. For example, the reaction of 1-fluoro-3-sulfinylacetone with allyl bromide upon deprotonation with LDA proceeded in 65% yield and 20% diastereomeric excess [63,64]. The promising reagent for creation of an asymmetric monofluoromethylene fragment is 1-(fluoromethyl)-2-[(*S*)-*p*-tolylsulfinyl]benzene **59** (Scheme 20). The *ortho*-sulfinyl group, which is distant from the reaction center, can simultaneously stabilize the lithium carbanion in the benzyl position and act as a chiral inductor. The interaction of sulfone **59** with primary allyl bromides in the presence of LDA proceeded at -78°C in less than a minute with the formation of homoallyl fluorides **60** in good yields. The stereoselectivity of this process was moderate, which contrasts with the data for the previously studied



Scheme 19. Co-catalyzed allylation reaction.

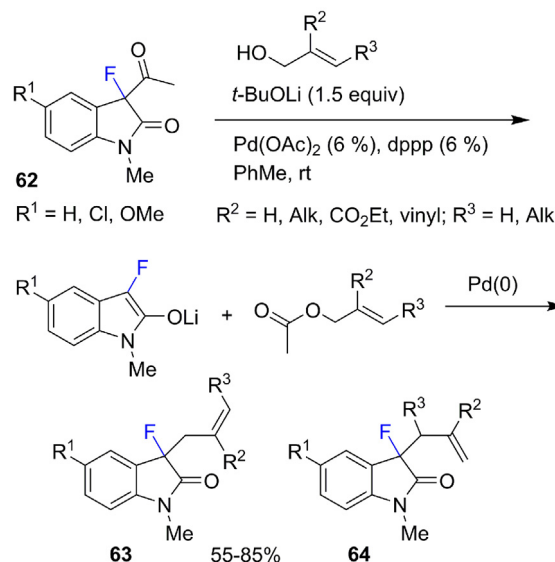
Scheme 20. Reactions of 1-(fluoromethyl)-2-[(*S*)-*p*-tolylsulfinyl]benzene.

reactions of sulfone **59** with carbonyl compounds, imines and Michael acceptors [65].

The interaction of sulfoxide **59** with propargyl halides upon treatment with LDA also proceeded regioselectively and with a high rate; however, the stereoselectivity of this process was significantly better. The diastereomeric excess of homopropargyl fluorides **61** in most examples exceeded 80%, with the exception of alkyne bearing a *para*-cyanophenyl substituent (de 30%). The authors explained this result by the presence of a stacking interaction between the triple bond and the aryl fragment, which affects the conformation of the active species. This interaction leads to the stabilization of the less favorable conformation and is more noticeable in the case of substrates with electron withdrawing substituents. The assumption is fully consistent with the data of quantum chemical calculations [65].

2.2. Reactions catalyzed by transition metal complexes

3-Acetyl-2-oxoindoles **62** can react with allyl alcohols under deacetylation conditions with the catalysis by palladium salts (Scheme 21). Using palladium acetate as the catalyst and 1,3-bis(



Scheme 21. Pd-catalyzed deacetylation allylation.

diphenylphosphino)propane as the ligand a number of the substitution products **63** and **64** were obtained with the yields from moderate to high. In the case of allyl alcohol bearing propyl substituent at the double bond ($R^3 = n\text{-Pr}$), the α -substitution product **63** was formed exclusively, while in the case of crotyl alcohol ($R^3 = \text{Me}$) and penta-1,4-diene-3-ol ($R^2 = \text{vinyl}$), the mixtures of products **63** and **64** were isolated in 1:1 and 2:1 ratio [36]. The reaction mechanism involves transfer of the acetic group from the oxoindole to the allylic alcohol followed by palladium(0) catalyzed allylation of α -fluoro-substituted lithium enolate.

Palladium catalysis was the most effective in the reactions with allylic acetates and carbonates. In the presence of chiral ligands the process proceeded with high enantioselectivity, the examples are presented in Table 1.

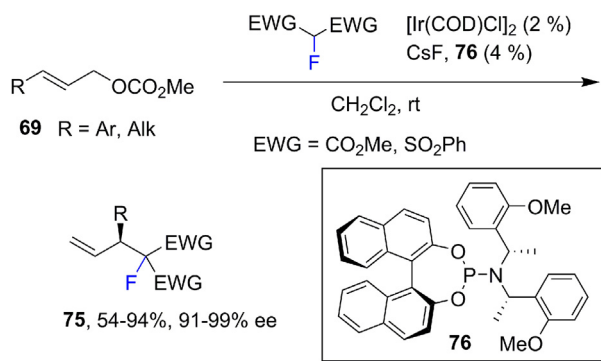
Asymmetric allylation was first demonstrated by the reaction of 1,3-diphenylallyl acetate **65a** with fluoromalonate **1** in the pres-

ence of phosphine ligand **67** bearing two prolinol fragments (entry 1). A combination of sodium acetate and bis(trimethylsilyl)acetamide was used as a base. As a result, the fluoromalonate (*R*)-**66a** was obtained in a quantitative yield and 95% enantiomeric excess [66]. In a similar reaction of acetate **65a** with malonate **1** in the presence of 2-(oxazoliny)pyridyl ligand (*R*)-Spymox another enantiomer (*S*)-**66a** was isolated in a moderate yield and excellent enantioselectivity (entry 2) [67].

A more general allylation protocol with a new type of S,N-ligand based on *tert*-butyl sulfinamide was demonstrated in 2015. Acetates **65** were transformed into malonates (*R*)-**66** in the presence of sulfinylamide catalyst **68** and using cesium carbonate as a base in high yields and good enantiomeric excess (ee 80 – 90%) (entry 3). The authors noted that 2-oxa-4-pentenyl substituent in the *ortho*-position of the pyridyl group crucially affected the enantioselectivity [68].

Table 1
Catalytic asymmetric fluoroalkylation reactions; allyl palladium chloride as catalyst.

Entry	Substrate	Product	Ligand	Lit.
1		 (<i>R</i>)- 66a , 99%, 95% ee		[66]
2		 (<i>S</i>)- 66a 58%, 99% ee		[67]
3		 (<i>R</i>)- 66 , 69–99%, 65–93% ee		[68]
4		 72 , 54–92%, 91–97% ee		[69]
5		 70 , 68 – 85% de > 95%, 92–99% ee		[70]
6	 $R^1, R^2, R^3 = \text{Alk}, \text{Ar}$ $X = \text{Br}, \text{OTf}$	 74 , 42–93%		[71]



Scheme 22. Iridium-catalyzed allylation reactions.

The reaction of acetates **65** with disulfones in the presence of S-PHOX ligand and cesium carbonate as a base proceeded in a similar manner (entry 4) [69].

Monosubstituted allyl carbonates **69** reacted with phosphoryl fluoroacetate in the presence of cesium carbonate with the formation of γ -substitution products **70** in good yields (entry 5). In the presence of ferrocenyl ligand **71** predominantly one diastereomer (de > 90%) with high enantioselectivity (ee up to 99%) was formed [70].

When bromides and triflates **73** bearing a diene fragment reacted with fluoromethyldisulfonate in the presence of potassium *tert*-butoxide, allenes **74** were formed (entry 6). The reaction proceeded exclusively as γ -substitution in good yields [71].

The reactions of stabilized monofluoroalkyl anions with allyl carbonates may also be catalyzed by iridium complexes. It was shown that the reaction of carbonates **69** with α -monofluorosubstituted malonates, acetoacetates and disulfones in the presence of 2 mol% of [Ir(COD)Cl]₂ and cesium fluoride as an activator easily proceeded at room temperature (Scheme 22). The reaction was also carried out in a stereoselective manner by the addition of phosphoramidite ligand **76**, which resulted in the formation of α -substitution products **75** in high yields and stereoselectivity [72,73].

3. Reactions of difluorinated carbanions

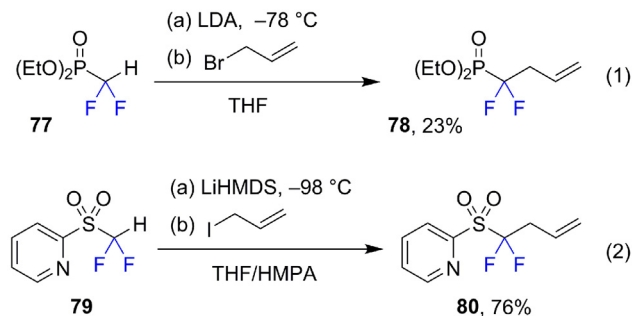
Geminal α,α -difluorinated carbanions generally have extremely low stability, and in the absence of charge delocalization decompose already at -78°C with the elimination of metal fluoride and formation of fluorocarbene [74]. Thus, nucleophilic reactions resulting in the transfer of RCF₂-fragment are possible only for the anions stabilized by electron-withdrawing groups (R = EWG), geminal difluorinated enolates, as well as activated silicon reagents and complexes of transition metals.

3.1. Reactions of carbanions stabilized by electron withdrawing groups

3.1.1. Organolithium compounds

The first example of the reaction of difluoromethylphosphonate **77** lithiated with LDA in tetrahydrofuran at -78°C with allyl bromide was demonstrated in 1982 (Scheme 23, equation 1). The yield of product **78** was only 23%, which was unexpectedly noticeably inferior to the results of the similar reaction of phosphonate **77** with other alkyl halides [75].

An example of the reaction of difluoromethyl 2-pyridylsulfone **79** with allyl iodide upon lithiation with LiHMDS was also demonstrated (Scheme 23, equation 2). The reaction product **80** was isolated in 76% yield. The authors noted that in the presence of an



Scheme 23. Allylation of difluoromethyl phosphonate and sulfone.

excess of base in the reaction mixture, β -elimination of the sulfinate group from olefin **80** occurred with the formation of the corresponding 1,1-difluorodiene [76].

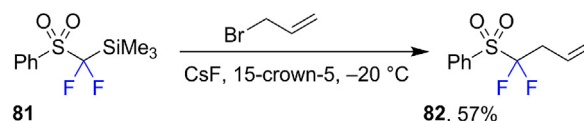
3.1.2. Activated silicon complexes

Reactions of α,α -difluoroalkyl substituted silanes with alkyl halides without transition metal catalysis are rare. The only example of such direct transformation is the alkylation of difluoro(phenylsulfonyl)methylsilane **81** with allyl bromide upon activation with cesium fluoride in the presence of two equivalents of crown ether. The product of this transformation, sulfone **82**, was isolated in 57% yield (Scheme 24) [77].

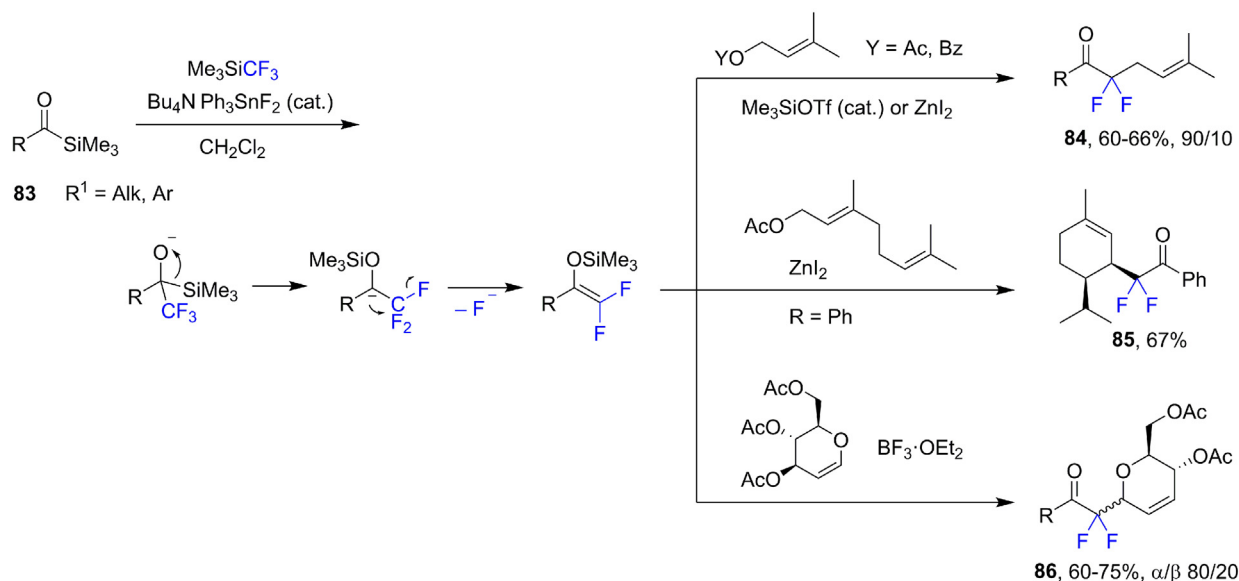
Geminal difluorinated silyl enol ethers have proved to be effective reagents for the synthesis of compounds with the CF₂-unit. One of the pathways of their *in situ* generation is the reaction of acylsilanes **83** with Me₃SiCF₃ (the Ruppert-Prakash reagent) in the presence of a catalytic amount of tetrabutylammonium difluorotriphenylstannate as a source of fluoride anion (Scheme 25). Silyl enol ether is formed after the Brook rearrangement of the initially formed CF₃-adduct followed by β -elimination of fluoride ion, which can activate another molecule of the Ruppert-Prakash reagent. The difluorinated silyl enol ethers were then reacted with prenylacetates and benzoates upon promotion by Lewis acids. Compounds **84** were obtained in this process in the form of a 90/10 mixture of regioisomers with a predominance of the expected α -substitution product [78].

In case of the reaction of the enol ether with geranyl acetate, the carbocyclic product **85** was formed. This was explained by the closure of the carbocationic intermediate followed by rearrangement. After reactions of various silyl enol ethers with an acylated glycol, the glycosylation products **86** were obtained regioselectively with 80/20 ratio of α - and β -anomers. The authors highlighted the strong dependence of the reaction efficiency on the choice of the Lewis acid [78].

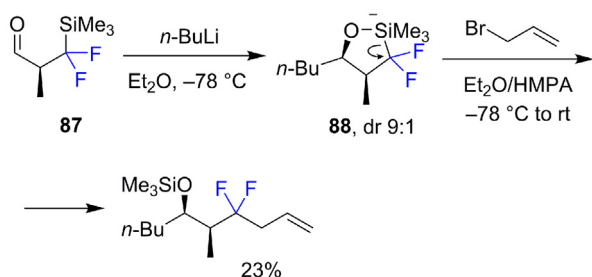
There is also a version of such transformation involving 1,4-Brook rearrangement. Thus, aldehyde **87** was treated with *n*-butyllithium with the formation of cyclic pentacoordinate silicon intermediate **88**, which was then directly reacted in presence of HMPA with various electrophiles, including allyl bromide (Scheme 26). As a result of this three-component coupling, the allylation product was isolated in a modest yield [79].



Scheme 24. Allylation of difluoro(phenylsulfonyl)methylsilane.



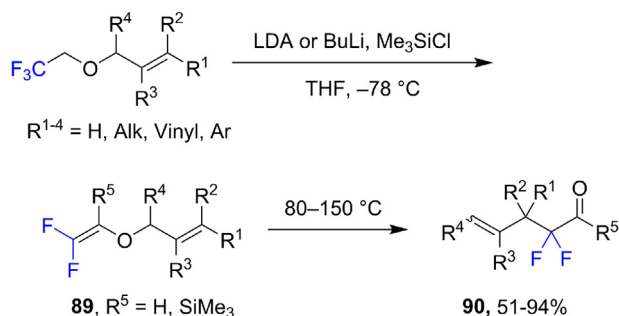
Scheme 25. Generation of difluorosilyl enol ethers and their allylation reactions.



Scheme 26. Tandem nucleophilic addition/1,4-Brook rearrangement/allylation reaction.

3.1.3. Claisen rearrangement

Claisen rearrangement of *gem*-difluorinated vinyl ethers leads to products, which formally correspond to the products of allylation of difluoroenolates. Thus, treatment of trifluoroethanol ethers with butyllithium or LDA affords *gem*-difluorovinyl ethers **89** by elimination of hydrogen fluoride [80] (Scheme 27). Ethers **89** undergo thermally-activated rearrangement into α,α -difluorosubstituted carbonyl compounds **90** in good yields. If two equivalents of the lithiating agent in the presence of chlorotrimethylsilane were used, the silyl derivatives (**89**, $R^5 = \text{SiMe}_3$) were formed, the rearrangement of which also led to the functionalized acylsilanes **90** [80,81]. A procedure for the generation of intermediates **89**



Scheme 27. Claisen rearrangement of (difluorovinyl)allyl esters.

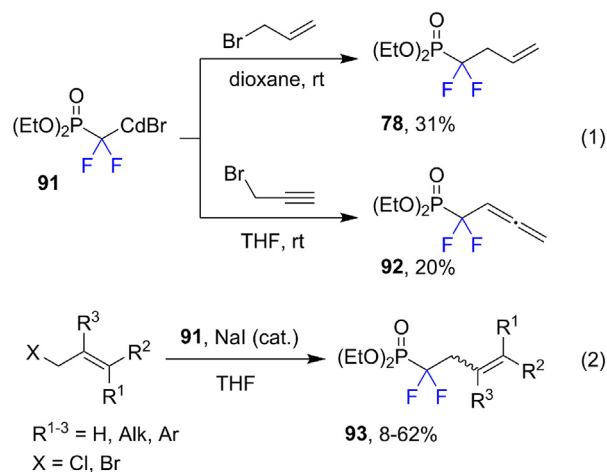
from α -fluorosulfoxides by aldol reaction and subsequent *O*-allylation was also reported [82].

3.2. Reactions catalyzed by transition metals

3.2.1. Cadmium reagents

Cadmium reagent **91** was synthesized for the first time in 1981 from diethyl (bromodifluoromethyl)phosphonate and cadmium metal (Scheme 28, equation 1). It was shown that the solution of compound **91** may be stored for a long time at room temperature due to its unique stability. According to ^{19}F and ^{31}P NMR monitoring, it was found that in solution reagent **91** always exists in a Schlenk equilibrium with its dimeric form. After the dioxane solution of this reagent was mixed with allylbromide at room temperature, product **78** was formed in 31% yield [83]. Later, in 1988, an uncatalyzed reaction of the complex **91** with propargyl bromide in tetrahydrofuran was also carried out. In the latter case, allene **92** was isolated in 20% yield, and the formation of the isomeric alkyne in the amount of 7% was observed [84].

In further studies it was shown that the reaction of reagent **91** with electrophiles may be improved by addition of a catalytic



Scheme 28. Reactions of cadmium diethyl difluoromethylphosphonate.

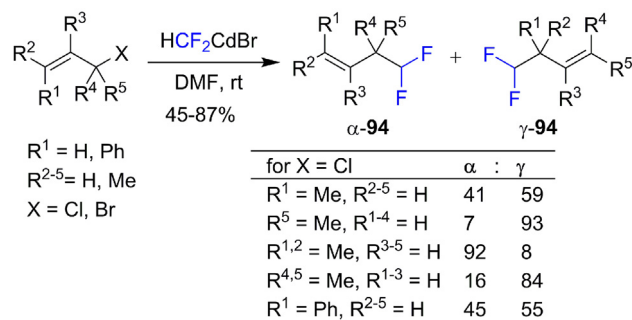
amount of sodium iodide. The products of the reaction of **91** with primary allyl halides were isolated as single regioisomers **93** in moderate yields (Scheme 28, equation 2) [84,85]. The main by-product in most cases was diethyl difluoromethylphosphonate (up to 57%), forming as a result of protonation of the cadmium reagent.

Interaction of bromodifluoromethane (HCF_2Br) and dibromodifluoromethane (CF_2Br_2) with cadmium metal in DMF allows to generate di- and trifluoromethylcadmium bromides respectively [86]. Non-catalytic coupling of CF_3CdBr with allyl bromide at room temperature was reported with the yield of 70%; however, the reaction rate was extremely low (several days). The reaction of HCF_2CdBr under similar conditions proceeded much faster and gave the allylation product with the yield of 85%. The reactions of difluoromethylcadmium in DMF with allylic substrates bearing various substituents were also studied in detail (Scheme 29) [87,88]. It was shown that the allylation products **94** formed as the mixtures of regioisomers with the overall yields of 45–87%. The selected examples illustrate that the nature of the competition between α - and γ -substitutions as shown in Scheme 28. In case of steric hindrances at the carbon bearing the halogen atom ($\text{R}^4/\text{R}^5 = \text{Me}$) a strong predominance of the γ -substitution product was observed. At the same time, when the double bond was sterically hindered ($\text{R}^1 = \text{Me}, \text{Ph}$), α - and γ -substitution products were formed in approximately equal proportions. The authors concluded by this fact that γ -substitution with 1,3-rearrangement of the alkene is a generally preferable pathway.

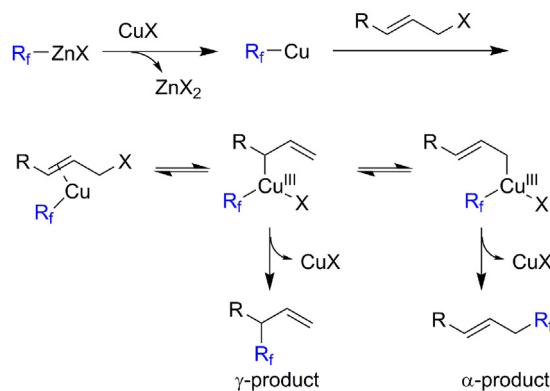
3.2.2. Copper reagents

Organocopper reagents demonstrate excellent activity in reactions with allylic electrophiles, but their limited stability in most cases makes them unsuitable for a long term storage. On the other hand, organofluorine zinc derivatives are quite stable and non-toxic, though their allylation processes are often unsatisfactorily slow. In this regard, transmetalation from zinc or cadmium to copper is a common method for generation of fluorinated copper species, which then couple efficiently with allyl halides (Scheme 30). It is believed that this processes proceed through the reversible formation of short-lived copper (III) complexes with their further transformation into the products of α - and γ -substitution by reductive elimination. The key factor is the rate of the equilibrium between the copper(III) complexes, which determines the regioselectivity of the reaction [89]. It also should be noted that the regeneration of copper (I) salts occurs during the reaction, so the process may be carried out in a catalytic version.

The regioselectivity of the reactions of fluorinated copper species with allyl and propargyl electrophiles is often quite complicated and difficult to predict, but several general rules can be formulated. If one of the positions, either α or γ , is free from substituents, then the linear products are formed selectively as *E*-



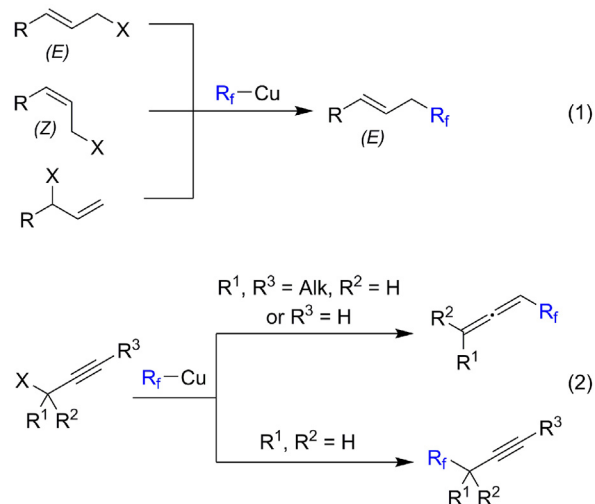
Scheme 29. Regioselectivity in allylation of difluoromethylcadmium reagent.



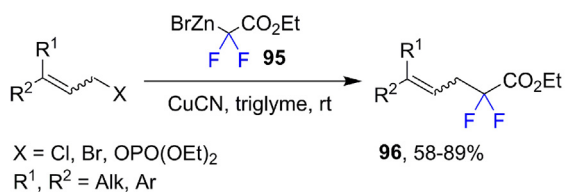
Scheme 30. Mechanism of the allylation of R_fCu .

isomers regardless of the configuration of the double bond in the substrate (Scheme 31, equation 1). If there are substituents in both α - and γ -positions, the mixture of linear and branched products is usually observed, and the ratio in this case strongly depends on the steric factors of the substituents and the ligands on copper. The presence of $\text{Pd}(0)$ in the reaction mixture can completely change the reaction outcome, as it will be shown on several examples below, because palladium tends to form stable η^3 -allylic complexes instead of σ -complexes. The regioselectivity in this case depends on whether the fluorinated nucleophile comes from the palladium atom or from the inner sphere.

In case of alkynes, the reaction with R_fCu is quite sensitive to the nature of the substituent near the triple bond (Scheme 31, equation 2). The reaction with terminal alkynes ($\text{R}^3 = \text{H}$) always proceeds as γ -substitution, providing the corresponding allenes regardless of the substituents R^1 and R^2 . In case of internal alkynes ($\text{R}^3 = \text{Alk}$) the formation of allenes is still preferable if one of the substituents R^1 or R^2 is also an alkyl group. If both substituents R^1 and R^2 are hydrogens, the α -substitution with the formation of the corresponding alkyne becomes the main pathway. In the other cases the mixtures of isomers arising from α - and γ -substitution are usually formed. It should be noted, however, that bulky substituents near the triple bond (such as TIPS) can completely block the γ -substitution pathway.



Scheme 31. General trends in the regioselectivity of the reactions of R_fCu with allyl and propargyl electrophiles.



Scheme 32. Alkylation of ethoxycarbonyldifluoromethyl zinc bromide.

The first example of the copper-mediated transmetalation/allylation process was demonstrated in 1998 on zinc difluoroacetate **95**, which was previously generated from ethyl bromodifluoroacetate and zinc metal in triglyme (Scheme 32). Reagent **95** interacted with primary allyl halides and phosphates in the presence of an equivalent amount of CuCN; as a result, α -substitution products **96** were obtained in good yields. In the presence of chlorotrimethylsilane, which is known for its ability to accelerate the reactions with non-fluorinated cuprates and to promote the formation of γ -substitution products, a significant decrease in yield was observed. Allyl acetates also were completely inert under these conditions. The yields of products **96** were slightly increased in the presence of catalytic amounts of a palladium complex $\text{PdCl}_2(\text{PPh}_3)_2$ [90]. Several examples of the allylation of the similar copper reagent prepared by the direct metalation of methyl iododifluoroacetate were reported [91,92].

Zinc organocuprate **97** may also be prepared from diethyl bromodifluoromethyl phosphonate by direct metalation with zinc followed by the addition of copper (I) bromide (Table 2). The reactions of cuprate **97** with 1,4-dibromobutene [93], 1-bromo-3-methylbut-2-ene [94], (3-bromoprop-1-yn-1-yl)trimethylsilane [95], 2-(bromomethyl)acrylic acid [96] and its ester [96,97] were described, and in all these cases the corresponding α -substitution products were formed.

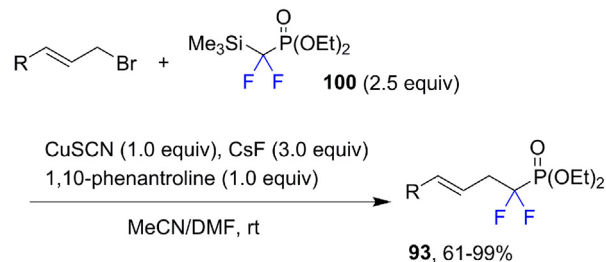
The interactions of the reagent **97** generated from bromodifluoromethyl phosphonate with propargyl electrophiles were studied in detail. Good activity was observed in the reactions with bromides, acetates and tosylates, while propargyl carbonates were found to be less active. In the absence of steric hindrances, the reaction proceeded predominantly as γ -substitution with the formation of allenes **98**. High yields and complete regioselectivity were achieved for propargyl bromide and tosylate by carrying out the reaction in dimethylformamide. Some examples of the interactions of reagent **97** with substituted propargyl halides are shown in Table 2 [98]. In the cases of steric hindrances within a tri-

ple bond, the formation of a significant amount of α -substitution products **99** was observed (entries 1,2) even with the complete reversal of regioselectivity (entry 8) [99]. A complete inhibition of the reaction when both reaction centers were blocked by bulky substituents was noted (entry 7).

Difluoro(trimethylsilyl)methylphosphonate **100** in the presence of basic activators can also act as a precursor of the corresponding cuprate (Scheme 33). In reactions of phosphonate **100** with various allyl bromides a number of α -substitution products **93** were obtained with the moderate to excellent yields. The authors reported a significant effect of 1,10-phenanthroline, which increased the yield of the process. The quantitative formation of the cuprate from the phosphonate **100** in the presence of cesium fluoride and copper thiocyanate was confirmed by ^{19}F NMR [100].

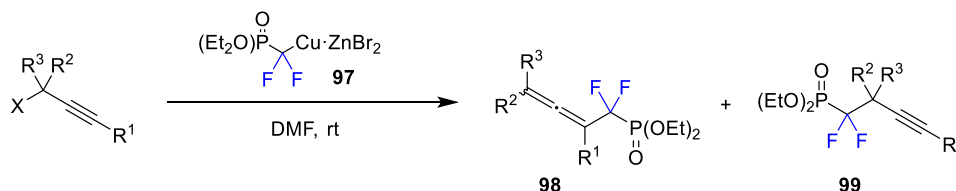
The possibility of a catalytic allylation of fluorinated organometallic reagents was shown for the first time in 1989 on the organozinc phosphonate **101** (Scheme 34). Unlike its cadmium analogue **91**, the reagent **101** interacts with allyl halides very slowly. At the same time, when a catalytic amount of copper (I) bromide was added, the reaction of **101** with allyl chlorides and bromides in glyme was complete within 12 h at room temperature. As a result, α - and γ -substitution products **93** were regioselectively formed in moderate yields, and the direction of the reaction was determined by the substituents R^4 and R^5 . Upon interaction of reagent **101** with a deuterium-substituted allyl chloride ($\text{R}^4, \text{R}^5 = \text{D}$), the products of α - and γ -substitution were isolated with a ratio of 29:71. In a similar reaction of **101** with propargyl chloride, allene **92** was obtained in 86% yield [101].

A modified version of this process allows to start the reaction directly from the diethyl bromodifluoromethylphosphonate **102**

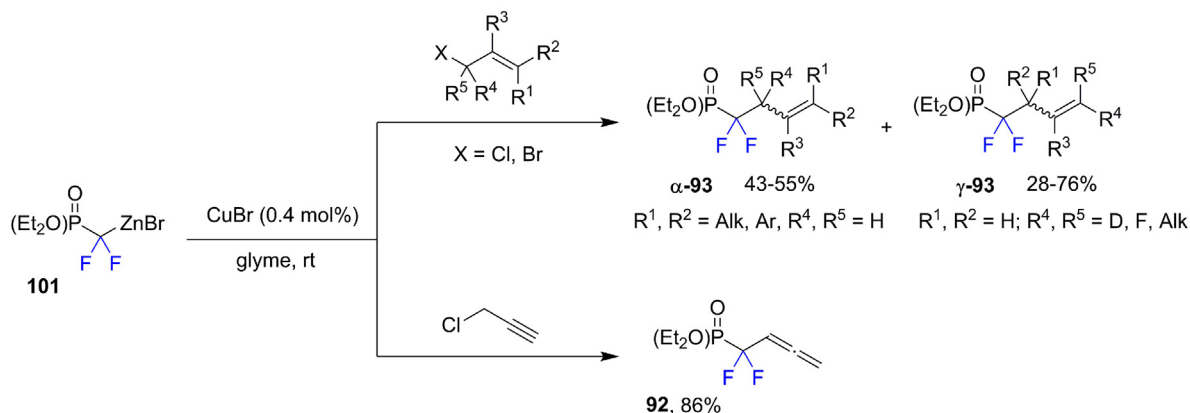


Scheme 33. Copper-promoted allylation of difluoro(trimethylsilyl)methylphosphonate.

Table 2
Couplings of reagent **97**.



Entry	R^1	R^2	R^3	X	98 : 99	Yield, %
1	Me	H	H	Br	23 : 77	71
2	Me	H	H	OTs	48 : 52	95
3	H	<i>i</i> -Pr	H	OTs	>99 : 1	98
4	H	CH_3	CH_3	OAc	>99 : 1	93
5	H	$(\text{CH}_2)_5$		OAc	>99 : 1	73
6	H	Ph	H	OAc	>99 : 1	74
7	Ph	Ph	H	OAc	n/a	–
8	SiMe_3	H	H	Br	< 1 : 99	70

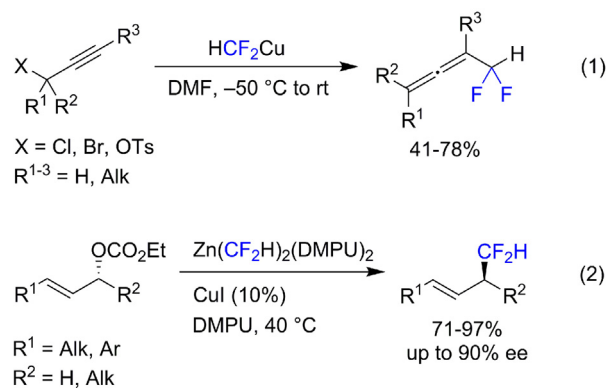


Scheme 34. Copper-catalyzed reactions of the zinc phosphonate.

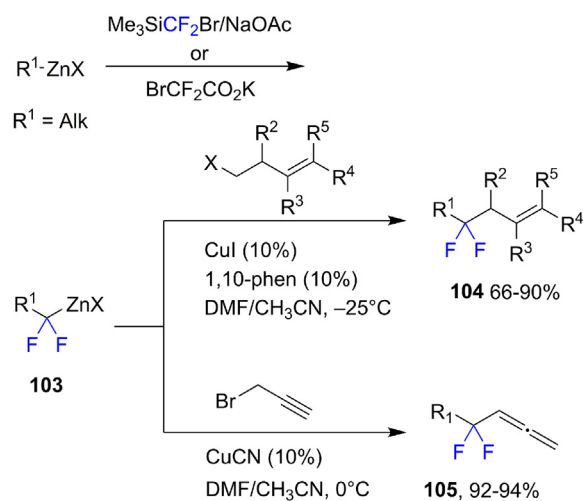
with no need to pre-generate the organometallic reagent in stoichiometric amounts (Scheme 35). In the reactions of bromide **102** with allyl halides in the presence of zinc dust and a catalytic amount of copper salt under mild conditions products **93** were isolated in moderate yields [102,103]. This process was implemented in the synthesis of one of the putative G-protein agonists GPR119 [104] and phosphoantigene HMBP [105].

Difluoromethylcopper derived from the corresponding cadmium reagent rapidly reacted with propargyl halides and tosylates with the regioselective formation of γ -substitution products in moderate to high yields (Scheme 34, equation 1). The reaction also proceeded with allyl halides, however, the selectivity of this process was quite low [88]. In a modified version of this process, a complex of bis(difluoromethyl)zinc with DMPU, which is a stable and isolable compound [106], reacted with allyl carbonates in the presence of catalytic amounts of copper iodide (Scheme 36, equation 2). The reaction proceeded as the selective α -substitution with the yields from good to excellent, and in case of secondary substrates, an inversion of stereochemistry was also observed (up to 90% ee). The authors explained such selectivity by steric factor, which affects the rate of reductive elimination step. Accordingly, a starting carbonate bearing two phenyl substituents ($\text{R}^1, \text{R}^2 = \text{Ph}$) was unreactive under these conditions [107].

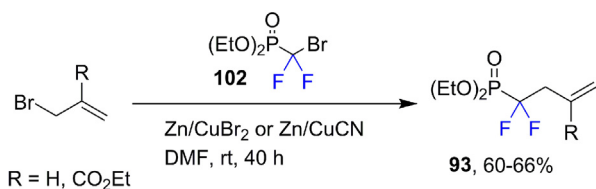
For a long time the chemistry of α,α -difluorosubstituted organometallic reagents was limited to the compounds containing exclusively electron withdrawing substituents. However, in 2013 it was shown that stable α,α -difluorinated organozinc reagents **103** may be obtained through the interaction of the organozinc compounds with the sources of difluorocarbene (Scheme 37). The most convenient sources of difluorocarbene turned out to be (bromodifluoromethyl)trimethylsilane ($\text{Me}_3\text{SiCF}_2\text{Br}$) [108] and potassium bromodifluoroacetate [109]. It was shown that reagents **103** can interact with various C-electrophiles with copper catalysis. The reaction of **103** with various allyl halides in the presence of the catalytic system involving copper iodide and 1,10-phenanthroline led to the α -substitution products **104** in high yields [109,110]. A similar reaction with propargyl bromide catalyzed by copper cy-



Scheme 36. Reactions of difluoromethylcopper.



Scheme 37. Copper-catalyzed reactions of difluorinated organozincs.



Scheme 35. Direct allylation of the bromodifluoromethylphosphonate.

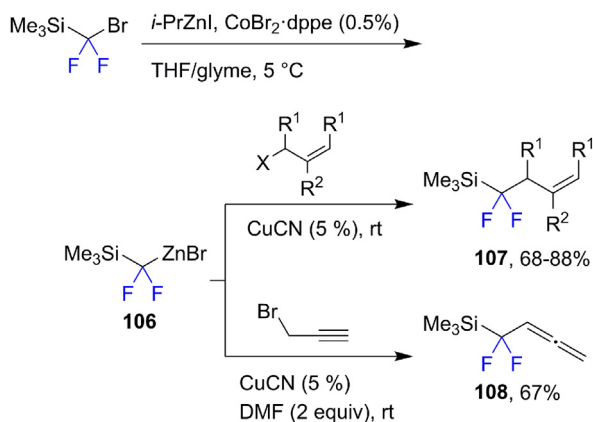
nide led to the formation of allenes **105** in excellent yields [111]. The presence of dimethylformamide in the reaction mixture noticeably increased the yield of the process, which was explained by an increase in the stability of the intermediate copper reagents. At the same time, the formation of cuprate from **103** was not reliably recorded, since in the presence of an equimolar amount of copper (I) salt the rapid decomposition of the fluorinated zinc reagent occurred.

A relatively stable bifunctional zinc reagent **106** was generated by the halogen-metal exchange reaction between (bromodifluoromethyl)silane and isopropylzinc iodide with $\text{CoBr}_2 \cdot \text{dppe}$ complex as the catalyst (Scheme 38). The structure of the reagent **106** was confirmed by an X-ray analysis. It should be noted that in this metalation process, the bromine atom from the CF_2Br -group is transferred into the ZnBr -fragment of the reagent **106**, which is quite unusual.

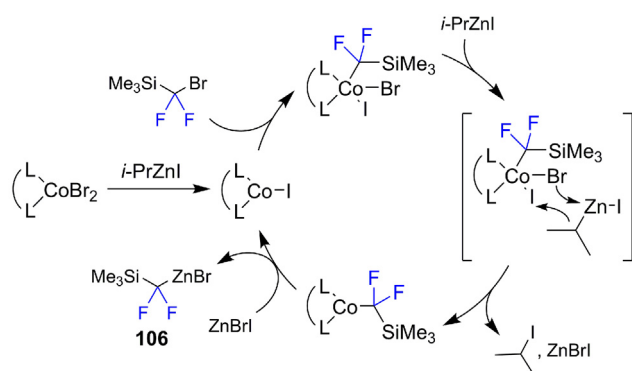
The mechanism accounting for this observation based on cobalt (I)/cobalt (III) cycle was proposed (Scheme 39). On the first step, cobalt (II)-phosphine complex is reduced by isopropylzinc iodide to give Co (I) species. The subsequent oxidative addition of the latter into the C-Br bond leads to Co (III) intermediate, which is reduced by another molecule of isopropylzinc iodide. The resulting Co (I) organometallic complex undergoes zinc/cobalt transmetalation providing reagent **106**. The formation of isopropyl iodide in this reaction was confirmed experimentally.

In the reaction of **106** with allyl halides catalyzed by CuCN, the substitution products **107** were obtained in good yields. α -Substitution proceeded selectively in most cases; however, in the reaction with methyl (*E*)-4-bromobut-2-enoate ($\text{R}^1, \text{R}^2, \text{R}^4 = \text{H}$; $\text{R}^3 = \text{CO}_2\text{Me}$) a mixture of α - and γ -substitution products was isolated with a 1:1 ratio [112]. In the same reaction of **106** with propargyl bromide, allene **108** was obtained in 67% yield, and the process worked well on a multi-gram scale [111].

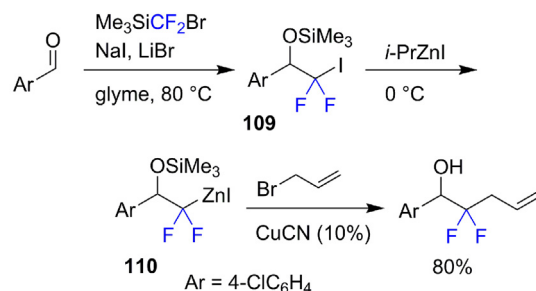
Trimethylsilyl ethers of iododifluoromethyl-substituted alcohols **109** are readily available by the reaction of aldehydes with (bromodifluoromethyl)silane in the presence of iodide anion (Scheme 40). The iodine atom in products **109** may be exchanged



Scheme 38. Consecutive metalation/allylation of (bromodifluoromethyl)silane.



Scheme 39. Mechanism of Co-mediated bromine/zinc exchange.



Scheme 40. Consecutive iododifluoromethylation/metalation/allylation reaction.

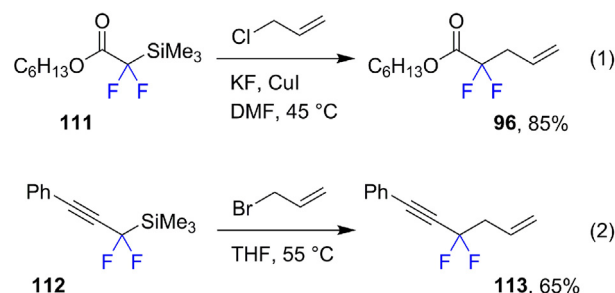
to zinc by a non-catalyzed interaction with isopropylzinc iodide, and the resulting difluoroorganozinc reagent **110** may react with electrophiles. The coupling of **110** with allyl bromide was promoted by copper cyanide, affording after desilylative work-up the corresponding alcohol in 80% yield based on starting aldehyde [113].

Fluorinated silicon reagents also may be used as the sources of fluorinated carbanions in copper-catalyzed allylation reactions. (Trimethylsilyl)difluoroacetate **111** and 1,1-difluoroprop-2-yn-1-yl silane **112** were reacted with allyl chloride and allyl bromide, respectively, in the presence of an equivalent amount of copper iodide upon activation with potassium fluoride (Scheme 41). As a result, the corresponding products **96** [114] and **113** [115,116] were obtained in good yields.

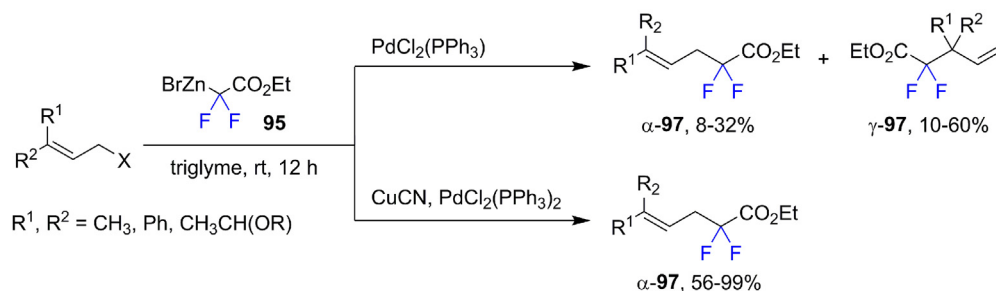
3.2.3. Palladium catalysis

A difference between the activity of the reagent **95** itself and its cuprate generated with stoichiometric amounts of copper cyanide was demonstrated in the allylation reaction of zinc difluorenoate **95** in the presence of a palladium catalyst (Scheme 42). In the case of pure zinc reagent **95**, the catalytic reaction gave products **97** in modest yields and low regioselectivity. The formation of γ -substitution products prevailed, which is a general rule for the reactions of this kind. In case of the generated organocopper reagent, the palladium-catalyzed reaction proceeded as selective α -substitution furnishing α -**97** products in high yields [90].

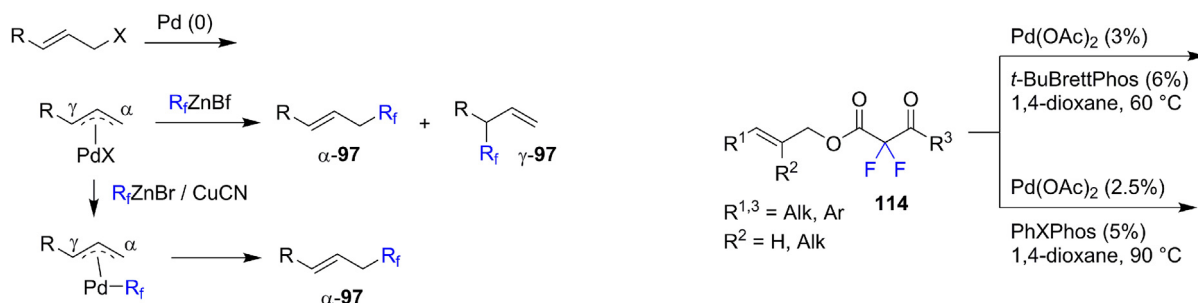
The authors suggested that the organocopper reagent formed by the zinc-copper exchange is more active and rapidly undergoes transmetalation with π -allyl palladium complex forming new mixed intermediate bearing the fluorinated fragment at the palladium atom, and this complex is then selectively converted into the α -substitution product α -**97** (Scheme 43). In case of weakly nucleophilic fluorinated zinc reagent $\text{BrZnCF}_2\text{CO}_2\text{Et}$, there is a competition between the attack at the γ -carbon atom and the attack at the palladium atom of the primary allylic intermediate, which results in the formation of the mixture of allylation products.



Scheme 41. Allylation of substituted difluoromethylsilanes.



Scheme 42. Palladium-catalyzed allylation of a zinc enolate.

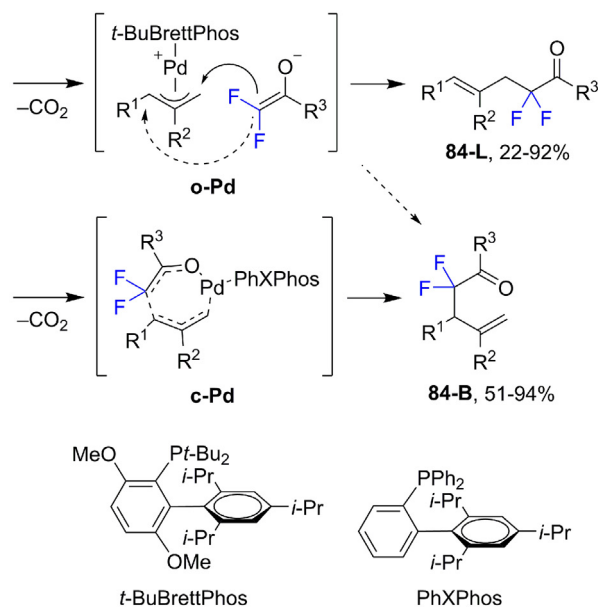
Scheme 43. Competition between α - and γ -substitutions.

The selectivity of the substitution may be controlled by ligands, which was shown in a Pd-catalyzed decarboxylative rearrangement of esters **114** (Scheme 44). In case of electron-rich and bulky ligand *t*-BuBrettPhos, the linear α,α -difluoroketones **84-L** were isolated in good yields and with moderate to high regioselectivity. Otherwise, with smaller and more electron-deficient PhXPhos the branched products **84-B** were obtained mostly in excellent yields and selectivities. The reaction likely proceeds through the formation of π -allyl palladium complex and carboxylate-anion, which undergoes rapid decarboxylation generating corresponding difluoroenolate. The authors suggested that in the first pathway, π -allyl palladium complex with bulky ligand **o-Pd** exists in an open form, and the process proceeds as an S_N2 -like attack at the less hindered position, where the competing process of S_N1 -like substitution at the secondary position, which is more electronically stabilized, is less preferable. This is supported by the fact that electron-rich substrates gave lower regioselectivity. In the second pathway, the palladacycle **c-Pd** may be formed, where the oxygen atom of the enolate is coordinated with palladium. This suggestion explains excellent regioselectivities for the branched substrates, and it also was supported by computational data, however, the experimental evidences of this pathway were not established [117].

4. Trifluoromethylation and perfluoroalkylation reactions

Like α,α -difluorosubstituted carbanions, perfluorinated carbanions in most cases are characterized by low stability [74,118] and they have no electron withdrawing groups capable of charge delocalization. For this reason, in reactions with C-electrophiles, synthetic equivalents of these carbanions or complexes with transition metals are mostly used.

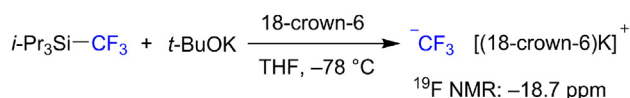
Long-term attempts to generate trifluoromethyl lithium or magnesium reagents in a stable form have been unsuccessful, which was explained by intrinsic instability of trifluoromethyl carbanion. However, later it was proved that the instability of anionic trifluoromethyl reagents is associated with Lewis acidic properties of the counter ions. Indeed, owing to the strong affinity of many metals to



Scheme 44. Ligand-controlled decarboxylative rearrangement.

fluorine, the metal cation such as lithium, magnesium, and even potassium can abstract the fluoride leading to rapid decomposition of the carbanion. A major breakthrough in this area was achieved in 2014 by Prakash, who demonstrated that if trifluoromethyl carbanion is devoid of interaction with metal counterions, which was achieved by application of strongly complexing crown ethers, it can be observed as a persistent species at low temperatures (Scheme 45) [119]. Moreover, in a tetrahydrofuran solution, the naked trifluoromethyl carbanion was stable at -78 °C for a few days, and only at temperatures from -50 to -35 °C its concentration decreased. These results clearly demonstrate the importance of effects of counterions on the behavior of carbanionic reagents.

Other perfluorinated alkyl lithium reagents were successfully prepared by means of halogen/lithium exchange or deprotonation reactions and used as versatile nucleophiles [74,118]. Perfluorinated aromatic compounds can also be easily generated. Pentafluoro-

**Scheme 45.** Generation of the CF₃-carbanion.

orophenyl lithium reagent can be used at low temperatures needed to suppress fluoride elimination with the formation of fluorinated dehydrobenzene. On the other hand, pentafluorophenyl magnesium bromide is stable at room temperature, and can be easily made from bromopentafluorobenzene and elemental magnesium according to a standard Grignard protocol.

4.1. Reactions of perfluorinated magnesium, cadmium and silver derivatives

Bromopentafluorophenylbenzene **115** may be transformed into a stable Grignard reagent upon treatment with magnesium in diethyl ether (Scheme 46, equation 1). Coupling of the Grignard reagent with allyl bromide afforded olefin **116** in high yield. The only by-product of this reaction was pentafluorobenzene, which could be a result of partial hydrolysis of the Grignard reagent [120]. Perfluorinated 1-bromodecane **117** also can interact with magnesium; however, a similar reaction of the corresponding Grignard reagent with allyl bromide resulted in the expected product **118** in only 11% yield (Scheme 46, equation 2). Most of by-products in this process were resulted from the hydrolysis of the magnesium reagent and the elimination of magnesium fluoride. There are few examples of allylation of perfluorinated cadmium and silver reagents **119a-c**, which are presented in Table 3.

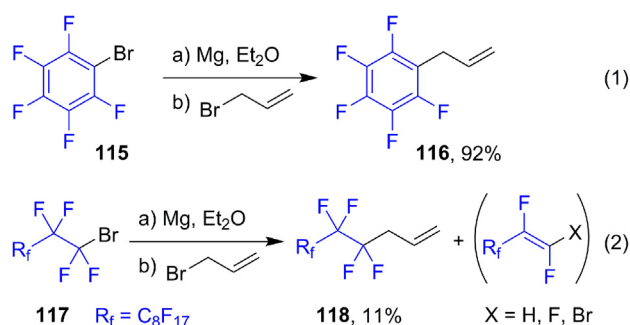
**Scheme 46.** Allylation of perfluorinated Grignard reagents.

Table 3
Allylation of cadmium and silver reagents.

Reagent	Product	Yield, %	Lit.
		78	[121]
		75	[122]
		50	[123]

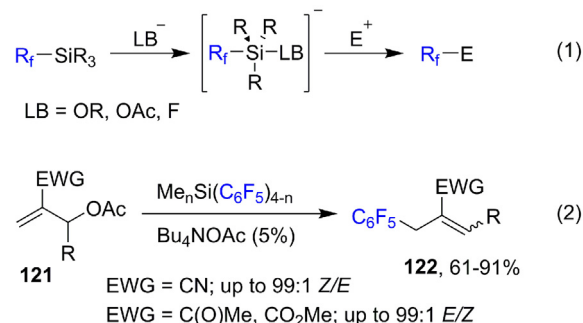
4.2. Reactions of perfluoroalkylsilicon derivatives

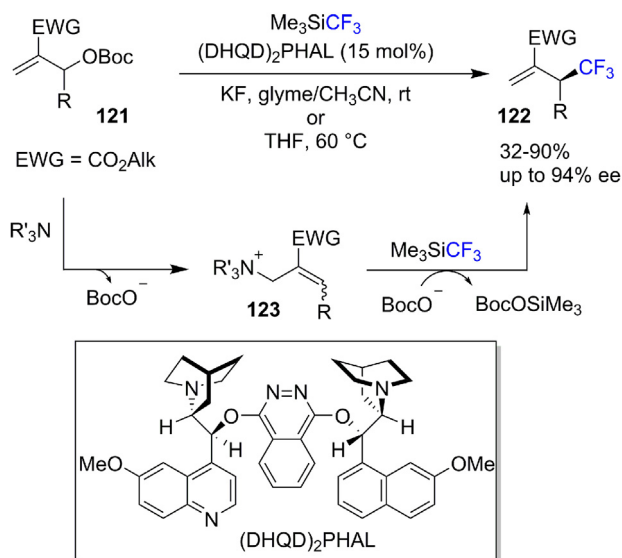
As it was highlighted in the section 3.1, organosilicon compounds bearing fluorinated substituents may be activated in reactions with electrophiles by various Lewis bases such as alkoxides, acetates or fluoride-anion via formation of pentacoordinated silicate complexes (Scheme 47, equation 1) (for details concerning Me₃SiCF₃, see section 4.4.3). In case of perfluorinated reagents, these silicates are too weak nucleophiles to interact with most of the allylating reagents. Nevertheless, in the presence of an additional acceptor group in the substrate such reactions may become possible.

Typically, the propensity towards formation of five-coordinate silicon species increases with the increase in number of electron withdrawing groups attached to silicon [124]. Silanes bearing three pentafluorophenyl groups were elaborated and used in reactions with electrophiles, and were shown to be highly susceptible to Lewis basic activation [125,126]. Thus, in reaction of the pentafluorophenyl-substituted silanes Me_nSi(C₆F₅)_{4-n} with the acylated Baylis-Hillman adducts **121** in the presence of an initiating amount of tetrabutylammonium acetate, selective γ-substitution was achieved with the formation of products **122** in high yields [127] (Scheme 47, equation 2). A similar transformation using the Ruppert-Prakash reagent proceeded in a significantly lower yield or led to the formation of by-products arising from the attack of the trifluoromethyl anion on the electron withdrawing carbonyl group [127].

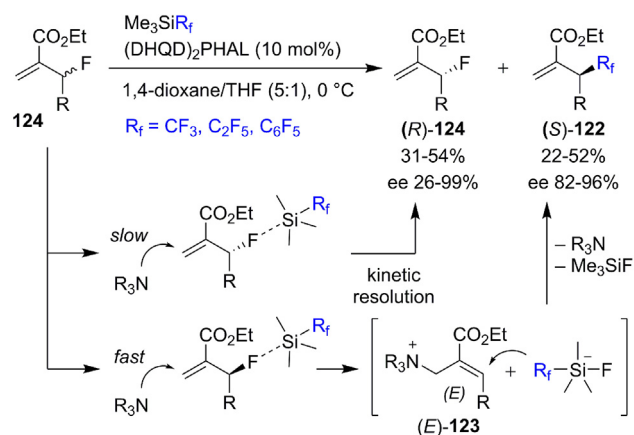
This problem was solved with the replacement of the acetate activator by quinuclidine or 1,4-diazabicyclo[2.2.2]octane (DABCO). S_N2'-Attack of the tertiary amine on the Boc-functionalized Baylis-Hillman adduct **121** leads to ammonium salt **123** and a carboxylate anion, which then acts as the silane activator (Scheme 48). The reaction of salt **123** with the Ruppert-Prakash reagent leads to another S_N2' attack with the formation of the product of formal α-substitution **122**. This process was carried out in an asymmetric version using chiral amine (DHQD)₂PHAL instead of quinuclidine. As a result, a number of trifluoromethylated products **122** were obtained in high enantioselectivities and moderate to high yields [128,129].

It is known that in the presence of chiral *Cinchona* alkaloids such as (DHQD)₂PHAL a kinetic resolution of Baylis-Hillman acetates and carbonates is possible [130]. The same strategy was developed for Baylis-Hillman allyl fluorides **124** under the conditions of enantioselective fluoroalkylation (Scheme 49). The reaction of racemic fluorides **124** with trifluoromethyl, pentafluoroethyl and pentafluorophenyl trimethylsilanes in the presence of (DHQD)₂-PHAL at slightly decreased temperatures resulted in the formation of fluoroalkylated alkenes (*S*)-**122** and the recovery of the initial (*R*)-isomers **124** with mostly good yields and excellent enantioselectivity.

**Scheme 47.** Reactions of acylated Baylis-Hillman adducts with C₆F₅-substituted silanes.



Scheme 48. Asymmetric reaction of Boc-substituted Baylis-Hillman adducts.



Scheme 49. Kinetic resolution of Baylis-Hillman fluorides by enantioselective trifluoromethylation.

lectivities. It was suggested that fluorinated silane Me₃SiR_f acts both as a source of fluorinated nucleophile and as an activator of the allylic fluoride, the latter activation is important due to high C-F bond energy. The reaction starts as S_N2'-nucleophilic attack of (DHQD)₂PHAL at the complex of racemic **124** with the silane, which presumably proceeds much faster with (*S*)-isomer than with (*R*)-isomer, resulting in the selective formation of intermediate (*E*)-**123**, while the compound (*R*)-**124** remains intact. Further interaction of (*E*)-**123** with the activated silicon complex leads to the formation of α -substitution product (*S*)-**122**. In the presence of another Cinchona alkaloid (DHQ)₂PYR, which has the reversed configuration at the quinuclidine centers, the reaction proceeded with a reversed enantioselectivity. The advantage of this method is an opportunity to produce two valuable compounds by one strike, given that the synthesis of chiral alkyl fluorides itself is a quite challenging task [131].

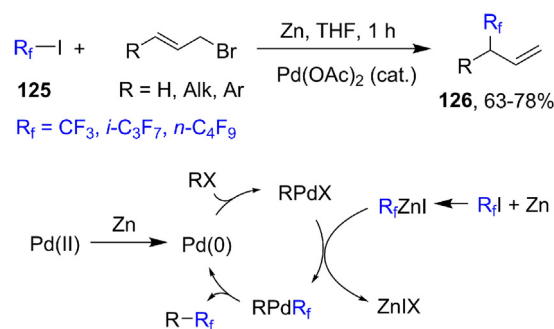
4.3. Reactions catalyzed by palladium

There are several examples of palladium-catalyzed reactions of perfluoroalkyl carbanions with allylic electrophiles. The coupling

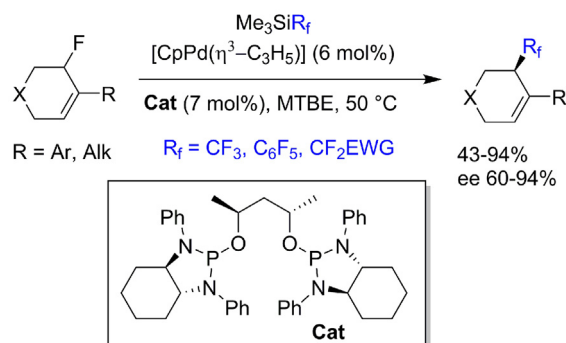
of perfluoroalkyl iodides **125** and primary allyl bromides upon treatment with zinc dust and palladium (II) acetate occurred within one hour at room temperature under ultrasonic conditions (Scheme 50). As a result, γ -substitution products **126** were isolated in good yields and high regioselectivities [132,133].

The reaction may also be catalyzed by palladium black with the yield of about 50%. The authors assumed that the real catalyst of the process is palladium (0) metal, which is formed after the reduction of the divalent salt with zinc dust. In the absence of ultrasonic irradiation, the reaction stops completely.

The asymmetric version of palladium-catalyzed fluoroalkylation of cyclic allyl fluorides with the different fluoroalkyl silicon reagents was also reported (Scheme 51). Fluorine showed the best results among the other conventional functionalities (acetate, carbonate, chloride) probably due to its ability to make complexes with fluorinated silanes (see section 4.2), which in a synergistic manner activates both a leaving group and a nucleophile. In the presence of diamidophosphite ligand the reaction proceeded smoothly with good yields and mostly excellent enantioselectivities. The reaction was also successfully employed on seven-membered rings, but failed with five-membered rings. The process showed good functional group tolerance, however, the presence of bromine substituent in the aromatic ring significantly decreased the reaction rate, which was explained by the competitive oxidative addition of palladium to the C-Br bond. When enantioenriched substrate was used, the retention of the configuration of the asymmetric center was observed, which means the reaction proceeds either by double inversion (ionization and subsequent nucleophilic addition) or by double retention mechanism. The authors proposed that the first pathway is preferable based on the literature data, however, the exact mechanism is not clear [134].



Scheme 50. Palladium-catalyzed allylation of perfluoroalkyl iodides.



Scheme 51. Palladium-catalyzed asymmetric fluoroalkylation of cyclic allyl fluorides.

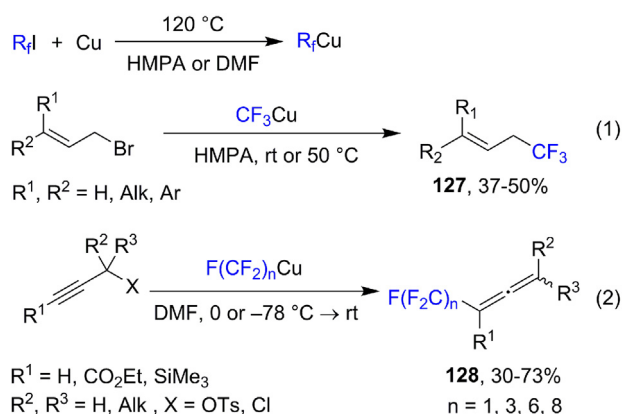
4.4. Reactions of copper (I) derivatives

Perfluoroalkyl derivatives of monovalent copper are widely used in organic synthesis to transfer fluorinated carbanions into various electrophiles. The organocopper compounds have excellent reactivity in allylation and propargylation processes and frequently provide high regioselectivities. At the same time, such species, with some rare exceptions, are prone to hydrolysis and decomposition via α -elimination pathway, and, therefore, they are not suitable for a long-term storage. To solve this problem, many methods of *in situ* generation of perfluoroalkyl copper complexes in stoichiometric or catalytic amounts were proposed [135–138]. Concerning trifluoromethyl derivatives, despite general formula of trifluoromethylcopper (CF_3Cu), the real structure of the reagent in most cases seems to be more complex and it obviously affects the reaction rate and selectivity [139]. Due to the high dependence of the properties of trifluoromethylcopper on the preparation protocol, the reactions in this section are grouped according to the ways of generation of the copper complex.

4.4.1. Reductive metalation

The reaction of copper metal with trifluoromethyl halides in polar donor solvents is one of the first methods for the preparation of trifluoromethyl copper (I) complexes. Trifluoromethyl copper may be generated from trifluoromethyl iodide and copper powder in HMPA at 120 °C, and the resulting reagent can interact with primary allyl bromides (Scheme 52, equation 1). As a result of such transformation, a number of α -substitution products were isolated in moderate yields [140]. A similar method of trifluoromethylation of allyl halides with bromotrifluoromethane was demonstrated with the use of electric current in the presence of a consumable copper anode. The stabilizing ligands such as phosphines or diamines were absolutely required for the successful formation of the organocopper species. Despite the high efficiency of the generation of trifluoromethyl copper (I) complexes, the allylation reaction under these conditions proceeded in low yields [141].

Long-chained perfluoroalkyl copper (I) complexes also may be prepared by the reaction of the corresponding iodides with copper powder at 120 °C in DMF. The cross-couplings of these organocopper species generated in this way with propargyl chlorides and tosylates were studied in detail (Scheme 52, equation 2). The process was quite rapid leading to the formation of allenes **128** with high regiospecificity. The products yields varied from moderate (for the longer chains ($\text{F}(\text{CF}_2)_n\text{Cu}$, $n = 6, 8$)) to good (for the shorter chains ($n = 1, 3$)) [142]. A similar reaction of perfluoroalkylcopper with propargyl bromides proceeded explosively and was not studied due to the significant danger and low efficiency [143].



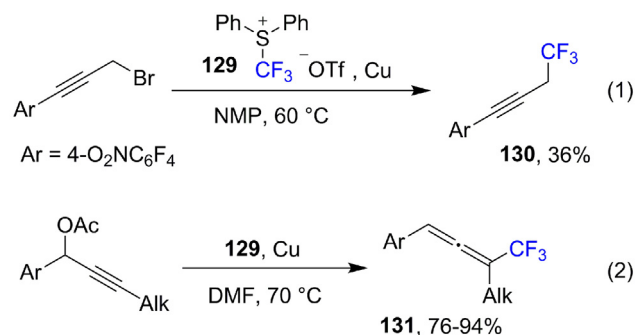
Scheme 52. Reaction of R_fCu prepared by direct metalation.

Diphenyl(trifluoromethyl)sulfonium triflate **129** is well-known for its activity in reactions of electrophilic trifluoromethylation (Scheme 53, equation 1). However, in the presence of copper powder it can form trifluoromethyl copper. The first example of propargylic trifluoromethylation with such system was demonstrated in 2011 on 1-(3-bromoprop-1-yn-1-yl)-4-nitrobenzene. The product of α -substitution **130** was isolated in 36% yield as a single regioisomer [144].

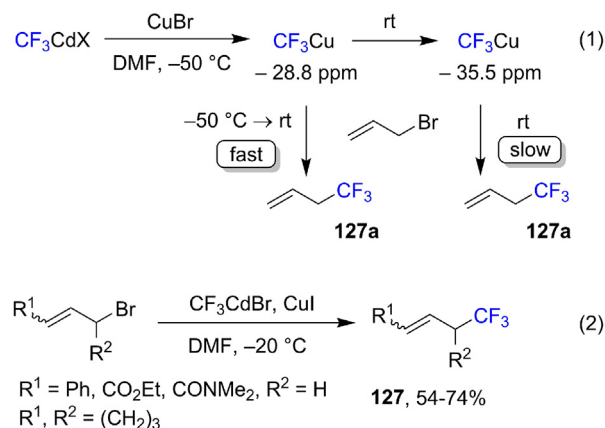
The similar process was studied in detail in 2014 on propargyl acetates (Scheme 53, equation 2). In the reaction of sulfonium salt **129** in the presence of copper powder, allenes **131** were isolated regioselectively and in high yields [145]. It was noted that this method of generation of copper reagents is the only effective way for the trifluoromethylation of propargyl acetates. Thus, the products **131** cannot be obtained under the conditions, which are used for the trifluoromethylation of propargyl halides and trifluoroacetates with $(\text{Ph}_3\text{P})_3\text{CuCF}_3$ [146] or the Ruppert-Prakash based system $\text{Me}_3\text{SiCF}_3/\text{KF}/\text{CuTC}$ [147].

4.4.2. Transmetalation of zinc and cadmium reagents

Perfluorinated copper compounds of various structures may be easily generated by metathesis of the corresponding zinc and cadmium reagents. For example, trifluoromethylcadmium bromide (CF_3CdBr) in DMF reacts with copper (I) bromide already at –50 °C forming CF_3Cu (Scheme 54, equation 1). Its reaction with allyl bromide while warming to room temperature proceeded rapidly with the formation of the substitution product **127a**. In the absence of allyl bromide at room temperature the originally formed species transformed into another trifluoromethyl copper species (detected by ^{19}F NMR), which was noticeably less active in the allylation



Scheme 53. Reactions of the sulfonium salt.



Scheme 54. Allylation of CF_3Cu prepared from CF_3CdX .

reaction [148]. After storage at room temperature, trifluoromethyl copper quantitatively disproportionated to copper (I) fluoride and a homologous complex of pentafluoroethyl copper $\text{CF}_3\text{CF}_2\text{Cu}$, which was not reactive with allyl bromide under the specified conditions.

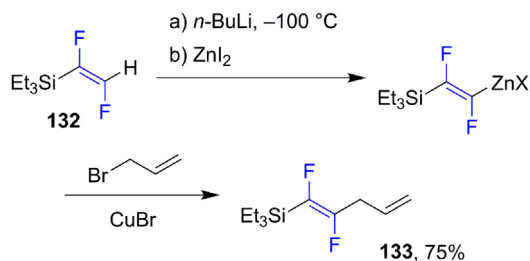
A combination of trifluoromethylcadmium bromide with copper iodide demonstrated high efficiency in the reactions with substituted primary allyl bromides and 3-bromocyclohexene, which was the first example of trifluoromethylation of allyl halides in mild conditions (Scheme 54, equation 2). The α -substitution products **127** were isolated with good yields and regioselectivity. A similar reaction was carried out with propargyl bromide, but in this case the γ -substitution product was isolated only in 43% yield [149].

There are several examples of cross-coupling reactions with fluoroalkyl substrates where metal center is on sp^2 hybrid carbon atom. Silane **132** was lithiated with further transmetalation with the formation of the corresponding zinc reagent, which readily entered into the copper-promoted reaction with allyl bromide affording product **133** in 75% yield (Scheme 55). [150]. The preparation of a similar vinyl organozinc reagent by the reaction of zinc metal with Z-1,2-difluoroiodoethylene was reported [151].

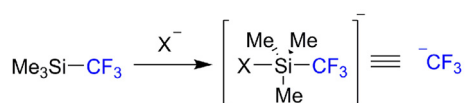
4.4.3. Transmetalation of activated silicon complexes

Silicon reagents are the most widely used equivalents of the trifluoromethyl carbanion. (Trifluoromethyl)trimethylsilane (Ruppert-Prakash reagent) is an air stable and widely available commercial compound, which becomes reactive only in the presence a Lewis basic activator [14,15]. Upon activation with a Lewis base, the five-coordinate siliconate species is generated [152–154] followed by transfer the trifluoromethyl carbanion towards an electrophile or a transition metal (Scheme 56). Typically, fluoride ion serves as a Lewis base, though other bases such as acetates and carbonates salts were successfully employed. The very facile basic activation of the Ruppert-Prakash reagent is apparently a consequence of electron withdrawing effect of three fluorine atoms. The analogous silane bearing only two fluorine atoms ($\text{Me}_3\text{SiCHF}_2$) is notably less reactive [155].

The first example of copper-promoted reaction of the Ruppert-Prakash reagent was demonstrated in 1991 involving cinnamyl bromide as electrophile [156]. A more detailed study of the interaction of this silane with allyl- and propargyl mesylates upon activation with potassium fluoride was published in 2013 (Scheme 57). It was shown that this process may be promoted by

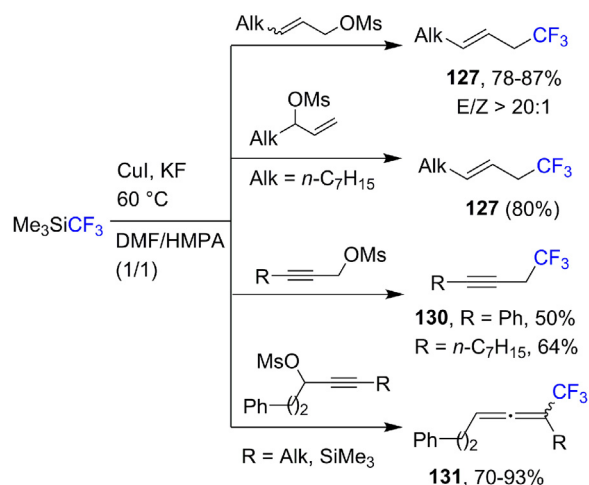


Scheme 55. Allylation of difluorovinylsilane.



Activators: Bu_4NF , Me_4NF , KF , AcONa , K_2CO_3

Scheme 56. Activation of the Ruppert-Prakash reagent.

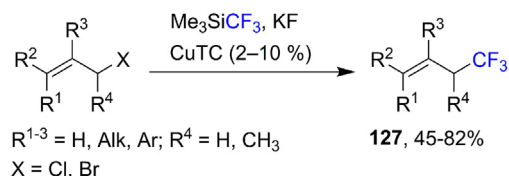


Scheme 57. Copper-mediated reactions of Me_3SiCF_3 with allyl and propargyl mesylates.

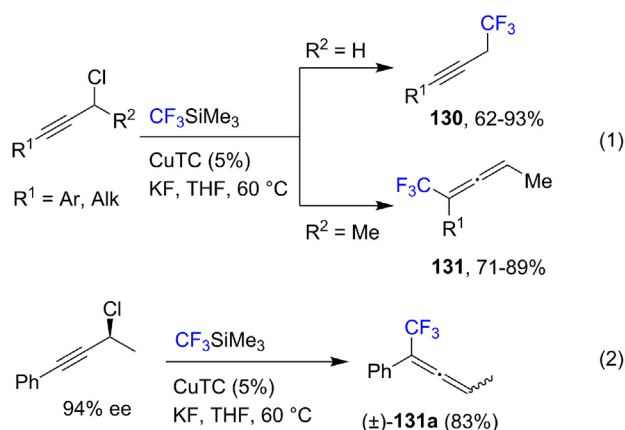
various copper (I) salts; however, the best results were achieved with the stoichiometric amounts of copper iodide. In these reactions with both primary and secondary mesylates, the linear substitution products **127** were formed corresponding to the attack of the nucleophile at the less sterically hindered position. Regardless of the configuration of the double bond in the substrates, product **127** was formed predominantly as *E*-isomer. A similar selectivity was observed for propargyl mesylates. The reaction of the silane with primary mesylates proceeded exclusively as α -substitution with the formation of alkynes **130**. At the same time, in case of secondary mesylates bearing substituents at both reaction centers, γ -substitution occurred regiospecifically with the formation of allenes **131** [157,158].

The catalytic versions of this transformation are possible. For example, in the reaction of Me_3SiCF_3 with allyl halides upon activation with potassium fluoride in the presence of 2 mol% of copper (I) thiophene-2-carboxylate (CuTC), the products of α -substitution were isolated (Scheme 58). The reaction with secondary substrates ($\text{R}^4 = \text{CH}_3$) required larger amounts of CuTC , while α -substitution products **127** were also regioselectively formed in moderate yields [159]. Also, an example of a similar reaction carried out in ionic liquid was reported [157].

Copper-catalyzed trifluoromethylation of propargyl chlorides with Me_3SiCF_3 was also performed under slightly different conditions (Scheme 59, equation 1). An increased amount of copper catalyst and higher temperature were required; however, the process proceeded in high yield and full regioselectivity. The direction of the substitution was determined by the steric factor; for example, in the reaction with primary chlorides only alkyne **130** was formed, while secondary chlorides were converted into allenes **131** in all cases. In the reaction of Me_3SiCF_3 with an optically pure alkyne racemic product ($\pm$)-**131a** was isolated (Scheme 59, equation 2). The authors suggested that this result indicates the formation of



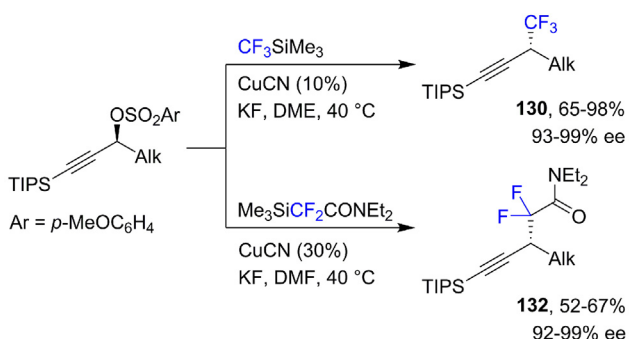
Scheme 58. Copper-catalyzed reaction of Me_3SiCF_3 with allyl halides.



Scheme 59. Copper-catalyzed reaction of Me_3SiCF_3 with propargyl chlorides.

the symmetric copper π -complex with the propargyl electrophile during the reaction [147].

On the other hand, the copper-catalyzed reaction of enantiomerically enriched secondary propargyl sulfonates with the Ruppert-Prakash reagent and (trimethylsilyl)difluoroacetamide proceeded as α -substitution with the inversion of configuration of the chiral center (Scheme 60). In both cases the products **130** and **132** were isolated in good or high yields with excellent regio- and stereoselectivity. The authors suggested that, unlike



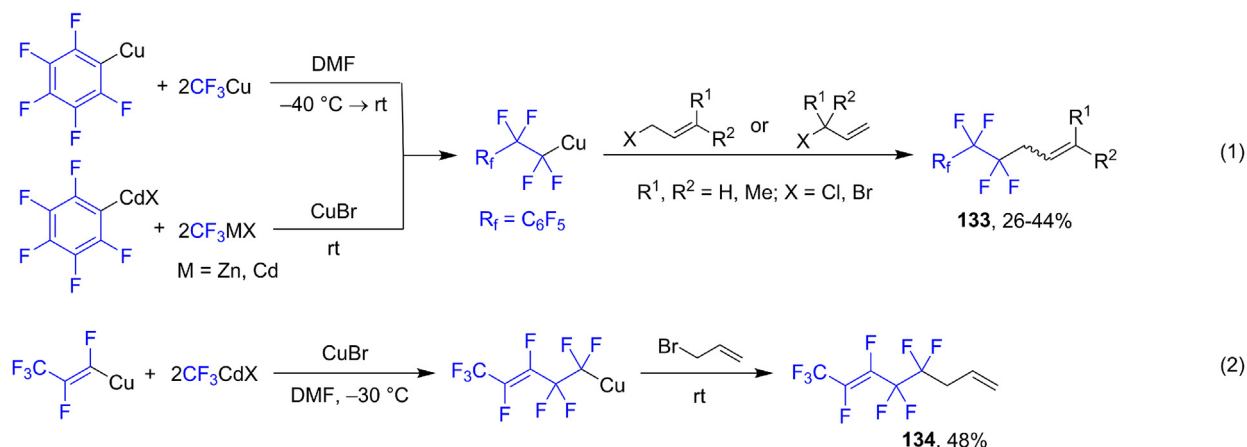
Scheme 60. Copper-catalyzed trifluoromethylation and difluoroamidation of propargyl sulfonates.

the case of propargyl chloride, the oxidative addition of the copper reagent to C-X bond ($\text{X} = \text{OSO}_2\text{Ar}$) happens as an $\text{S}_{\text{N}}2$ -substitution and causes no racemization, while bulky triisopropylsilyl (TIPS) group blocks the attack on γ -position. Unfortunately, allyl sulfonates failed to provide the corresponding fluoroalkylated products because of the instability of the substrates [160].

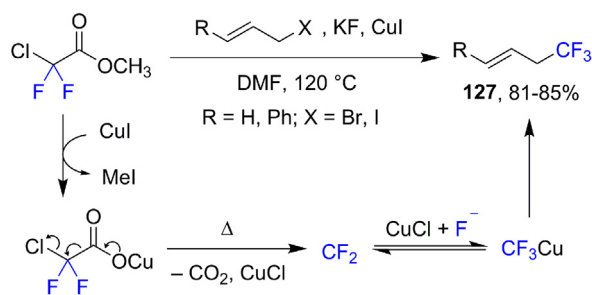
4.4.4. Synthesis from the sources of difluorocarbene

It was shown that difluorocarbene can perform double insertion into pentafluorophenylcopper with the formation of the corresponding perfluorinated copper reagent (Scheme 61, equation 1). Here, trifluoromethyl copper complex was used as a source of difluorocarbene. The same result was achieved in the reaction of pentafluorophenylcadmium with CF_3ZnX or CF_3CdX in the presence of copper salts [161]. The reaction of $\text{C}_6\text{F}_5\text{CF}_2\text{CF}_2\text{Cu}$ with primary, secondary and tertiary allyl halides proceeded regioselectively with the formation of only linear products **133** in moderate yields on two steps. The direction of the nucleophilic attack was completely determined by the position of the substituents in the halide. The process was also extended to (*Z*)-pentafluoropropenyl copper; the yield of the product **134** after the steps of insertion and allylation was 48% (Scheme 61, equation 2) [162]. It should be noted that in recent years, several novel convenient protocols for the generation of fluorinated organocopper species using Me_3SiCF_3 as a source of difluorocarbene were developed [163–165].

The relatively low stability of ligandless trifluoromethyl copper usually limits the possibility of using high temperatures or strong Lewis acids in the reactions of nucleophilic trifluoromethylation. At the same time, the protocols of generation of trifluoromethyl copper from a copper salt, fluoride anion, and a source of difluorocarbene right in the reaction mixture are free of such disadvantages. In these cases, trifluoromethyl copper always stays in equilibrium with the initial components, so the new active species is generated while the previous is consumed. The first example of trifluoromethylation of allyl halides under these conditions was achieved with methyl chlorodifluoroacetate in the presence of potassium fluoride and copper iodide upon heating to 120 °C. Methyl chlorodifluoroacetate undergoes demethylation with the formation of the corresponding copper salt, which gradually decomposes upon heating to copper chloride and difluorocarbene with the release of carbon dioxide (Scheme 62). The reaction furnished α -substitution products **127** in good yields [166]. This process was also performed under milder conditions with other sources of difluorocarbene, for example, potassium 3-oxa- ω -fluorosulfonyl



Scheme 61. Allylation of copper reagents prepared by double difluorocarbene insertion.



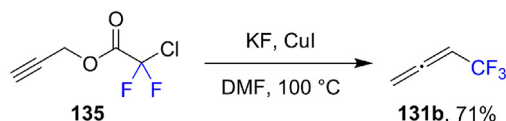
Scheme 62. Copper-mediated reaction of methyl chlorodifluoroacetate.

pentanoate ($\text{FSO}_2\text{CF}_2\text{CF}_2\text{OCF}_2\text{CO}_2\text{K}$), which readily decomposes already at 45 °C [167,168].

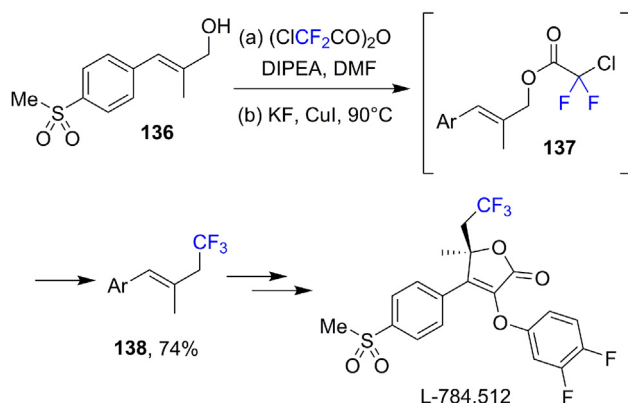
In a similar transformation of propargyl ether **135**, chlorodifluoroacetate acts not only as a source of difluorocarbene, but also as a leaving group (Scheme 63). Chlorodifluoroacetate-anion is released as the substitution reaction proceeds and then decomposes at high temperature releasing difluorocarbene. After this transformation, allene **131b** was isolated with 71% yield [169].

This approach allows the direct conversion of allyl and propargyl alcohols to trifluoroethyl-substituted alkenes and alkynes, which was demonstrated in the synthesis of the cyclooxygenase inhibitor L-784,512 (Scheme 64). Alcohol **136** was converted into the trifluoromethylated product **138** in one step in 74% yield without isolation of the intermediate **137**. An alternative transformation of the alcohol **136** to the corresponding allyl iodide with its further trifluoromethylation by potassium chlorodifluoroacetate/KF/CuI system yielded product **138** only in 43% [170].

This process was also made in a catalytic version and at lower temperatures with the allyl and propargyl derivatives of bromodifluoroacetate. Sodium bromodifluoroacetate was used as an initiator of this reaction since it readily decomposes upon heating at 50 °C with the formation of difluorocarbene. Its subsequent interaction with copper salt, fluoride-anion and ester **137** leads to the



Scheme 63. Reaction of propargyl chlorodifluoroacetate.



Scheme 64. Combination of chlorodifluoroacylation and trifluoromethylation reactions.

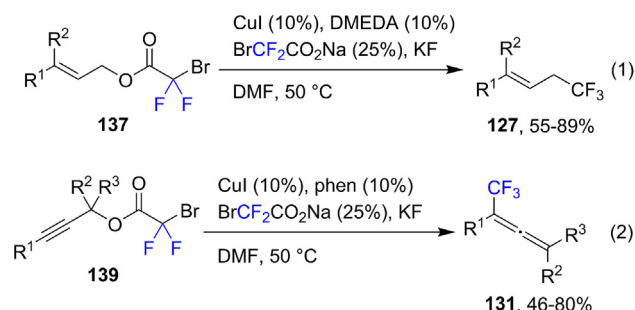
formation of α -substitution products and regeneration of bromodifluoroacetate anion (Scheme 65, equation 1). The reaction proceeds in good yields and excellent *E/Z*-selectivity [171]. The authors noted that preheating of the mixture of copper iodide, *N,N'*-dimethylethylenediamide (DMEDA), sodium bromodifluoroacetate and potassium fluoride in DMF for ten minutes before the addition of bromodifluoroacetate allowed to increase the yield by 15–20%. The similar reaction with propargyl derivatives proceeded as α -substitution with rather low selectivity, as the ratio of products varied from 1.2:1 to 4.0:1 [172]. However, under slightly modified conditions using 1,10-phenantroline as an additive, the reaction of propargyl esters **139** proceeded exclusively as γ -substitution, furnishing allenes **131** in good yields (Scheme 65, equation 2) [173]. Both allylation and allenation reactions with this system were later performed as a one-step deoxotrifluoromethylation of the corresponding alcohols, where the intermediate bromodifluoroacetic ester was generated *in situ* [174].

4.4.5. Isolated copper complexes

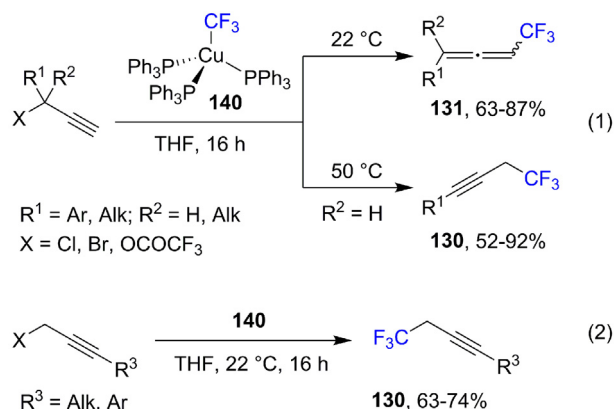
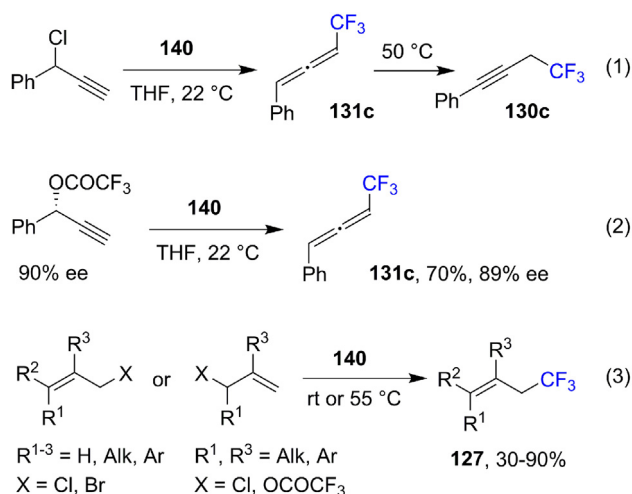
The advantage of utilization of trifluoromethylcopper complexes in the isolated state is a possibility to skip preparation steps, which allows to expand the range of reaction conditions. However, such complexes are quite rare. In 2012 the first example of a selective trifluoromethylation of propargyl halides and trifluoroacetates with the shelf-stable stable trifluoromethyl copper complex $(\text{Ph}_3\text{P})_3\text{CuCF}_3$ (**140**) for was reported (Scheme 66, equation 1) [146]. This reagent was previously synthesized from copper (II) fluoride and (trifluoromethyl)trimethylsilane, isolated and fully characterized [175]. It was found, that the selectivity of the interaction of complex **140** with terminal secondary and tertiary propargyl halides and trifluoroacetates was temperature-dependent; as a result, the series of allenes **131** and the isomeric products of their rearrangement **130** were obtained in good yields and selectivities. In the reactions of **140** with internal primary propargyl halides, α -substitution products **130** were also formed in good yields (Scheme 59, equation 2).

The authors suggested that the unusual temperature dependence of the selectivity is a result of a high lability of allenes **131** under the reaction conditions. On the example of (1-chloroprop-2-yn-1-yl)benzene they demonstrated that the γ -substitution product **131c** was formed at room temperature and quantitatively rearranged into the alkyne **130c** after the reaction mixture was heated to 50 °C (Scheme 67, equation 1). At the same time, the isolated allene **131c** did not undergo the rearrangement at 50 °C in THF even in the presence of $\text{CuCl}/\text{CuCl}_2$, copper (I) trifluoroacetate or reagent **140** [146].

Trifluoromethylation of propargyl trifluoroacetates with $(\text{Ph}_3\text{P})_3\text{CuCF}_3$ is of special interest, because, despite their low stability, these substrates may be generated *in situ* from readily available propargyl alcohols. The interaction of $(\text{Ph}_3\text{P})_3\text{CuCF}_3$ with prepared



Scheme 65. Copper-catalyzed trifluoromethylation of allyl and propargyl bromodifluoroacetates.

Scheme 66. Reactions of propargyl halides and trifluoroacetates with $(\text{Ph}_3\text{P})_3\text{CuCF}_3$.Scheme 67. Selectivity of the reactions of propargylic substrates with $(\text{Ph}_3\text{P})_3\text{CuCF}_3$.

this way chiral trifluoroacetate led to the formation of the allene **131c** with high stereochemical induction (Scheme 67, equation 2) [146]. A similar method was used for allyl halides and trifluoroacetates (Scheme 67, equation 3). Allyl chlorides and bromides reacted with $(\text{Ph}_3\text{P})_3\text{CuCF}_3$ at room temperature, but trifluoromethylation of trifluoroacetates required 55 °C. In case of both primary and secondary substrates, the linear products **127** were obtained in good yields. The authors noted that the reaction rate increased when the double bond had aromatic substituents with electron-donating groups [176].

5. Conclusion

Allyl and propargyl electrophiles are very active substrates in nucleophilic fluoroalkylation reactions. The protocols for the transfer of perfluorinated groups in the presence of transition metals, as well as the reactions of monofluorine-substituted methylene anions stabilized by one or two electron withdrawing substituents, are now developed quite well. The same processes for difluoromethylated carbanions are known predominantly for reagents bearing one electron withdrawing substituent; however, several methods of allylation and allenylation of unstabilized α,α -difluoroorganozinc compounds give an access to more complex structures. At the same time, reactions of allyl and propargyl substrates

with unstabilized monofluorine-substituted carbanions are practically unknown.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] K. Uneyama, *Organofluorine Chemistry*, Blackwell Publishing, Oxford, UK, 2006.
- [2] P. Kirsch, *M. Fluoroorganic Chemistry* (2004).
- [3] M. Inoue, Y. Sumii, N. Shibata, Contribution of organofluorine compounds to pharmaceuticals, *ACS Omega* 5 (19) (2020) 10633–10640, <https://doi.org/10.1021/acsomega.0c00830>.
- [4] J. Han, A.M. Remete, L.S. Dobson, L. Kiss, K. Izawa, H. Moriaki, V.A. Soloshonok, D. O'Hagan, Next generation organofluorine containing blockbuster drugs, *J. Fluorine Chem.* 239 (2020) 109639, <https://doi.org/10.1016/j.jfluchem.2020.109639>.
- [5] J. Wang, M. Sánchez-Roselló, J.L. Aceña, C. del Pozo, A.E. Sorochinsky, S. Fustero, V.A. Soloshonok, H. Liu, Fluorine in pharmaceutical industry: fluorine-containing drugs introduced to the market in the last decade (2001–2011), *Chem. Rev.* 114 (4) (2014) 2432–2506, <https://doi.org/10.1021/cr4002879>.
- [6] Y. Ogawa, E. Tokunaga, O. Kobayashi, K. Hirai, N. Shibata, Current contributions of organofluorine compounds to the agrochemical industry, *iScience* 23 (9) (2020) 101467, <https://doi.org/10.1016/j.isci.2020.101467>.
- [7] T. Fujiwara, D. O'Hagan, Successful fluorine-containing herbicide agrochemicals, *J. Fluorine Chem.* 167 (2014) 16–29, <https://doi.org/10.1016/j.jfluchem.2014.06.014>.
- [8] T. Okazoe, Overview on the history of organofluorine chemistry from the viewpoint of material industry, *Proc. Jpn. Acad., Ser. B* 85 (8) (2009) 276–289, <https://doi.org/10.2183/pjab.85.276>.
- [9] J. Gardiner, Fluoropolymers: origin, production, and industrial and commercial applications, *Aust. J. Chem.* 68 (1) (2015) 13.
- [10] J.-A. Ma, D. Cahard, Strategies for nucleophilic, electrophilic, and radical trifluoromethylations, *J. Fluorine Chem.* 128 (9) (2007) 975–996, <https://doi.org/10.1016/j.jfluchem.2007.04.026>.
- [11] T. Liang, C.N. Neumann, T. Ritter, Introduction of fluorine and fluorine-containing functional groups, *Angew. Chem. Int. Ed.* 52 (32) (2013) 8214–8264, <https://doi.org/10.1002/anie.201206566>.
- [12] P. Beier, M. Zibinsky, G.K.S. Prakash, Nucleophilic additions of perfluoroalkyl groups, in: *Org. React.*, John Wiley & Sons Inc, Hoboken, NJ, USA, 2016, pp. 1–492.
- [13] S. Barata-Vallejo, M.R. Torviso, B. Lantano, S.M. Bonesi, A. Postigo, Nucleophilic non-metal assisted trifluoromethylation and perfluoroalkylation reactions of organic substrates, *J. Fluorine Chem.* 161 (2014) 134–141, <https://doi.org/10.1016/j.jfluchem.2014.01.016>.
- [14] G.K.S. Prakash, A.K. Yudin, Perfluoroalkylation with organosilicon reagents, *Chem. Rev.* 97 (3) (1997) 757–786, <https://doi.org/10.1021/cr9408991>.
- [15] X. Liu, C. Xu, M. Wang, Q. Liu, Trifluoromethyltrimethylsilane: nucleophilic trifluoromethylation and beyond, *Chem. Rev.* 115 (2) (2015) 683–730, <https://doi.org/10.1021/cr400473a>.
- [16] G.K.S. Prakash, Z. Zhang, 11 - New Nucleophilic Fluoroalkylations A2 - Groult, Henri, in: H. Groult, F.R. Leroux, A. Tressaud (Eds.), *Modern Synthesis Processes and Reactivity of Fluorinated Compounds*, Elsevier, Amsterdam, 2017, pp. 289–337.
- [17] G. Li, C. Zhang, C. Song, Y. Ma, Progress in copper-catalyzed trifluoromethylation, *Beilstein J. Org. Chem.* 14 (2018) 155–181, <https://doi.org/10.3762/bjoc.14.11>.
- [18] C. Ni, J. Hu, The unique fluorine effects in organic reactions: recent facts and insights into fluoroalkylations, *Chem. Soc. Rev.* 45 (20) (2016) 5441–5454.
- [19] F.M. Bickelhaupt, H. Hermann, L. G. Boche, α -Stabilization of Carbanions: Fluorine Is More Effective than the Heavier Halogens, *Angew. Chem. Int. Ed.* 45 (2006) 823–826, doi:10.1002/anie.201605616.
- [20] Z. Zhang, A. Puente, F. Wang, M. Rahm, Y. Mei, H. Mayr, G.K.S. Prakash, The nucleophilicity of persistent α -monofluoromethyl anions, *Angew. Chem. Int. Ed.* 55 (2016) 12845–12849, <https://doi.org/10.1002/anie.201605616>.
- [21] L.A. Kaplan, H.B. Pickard, The α -fluorine effect and carbanion nucleophilicity, *Chem. Commun.* 0 (24) (1969) 1500–1501, <https://doi.org/10.1039/C29690001500>.
- [22] C. Ni, L. Zhang, J. Hu, Nucleophilic fluoroalkylation of α , β -enones, arynes, and activated alkynes with fluorinated sulfones: probing the hard/soft nature of fluorinated carbanions, *J. Org. Chem.* 73 (15) (2008) 5699–5713, <https://doi.org/10.1021/jo702479z>.
- [23] G.K.S. Prakash, F. Wang, N. Shao, T. Mathew, G. Rasul, R. Haiges, T. Stewart, G. Olah, A Persistent α -fluorocarbanion and its analogues: preparation, characterization, and computational study, *Angew. Chem. Int. Ed.* 48 (29) (2009) 5358–5362, <https://doi.org/10.1002/anie.200901414>.
- [24] X. Shen, L. Zhang, Y. Zhao, L. Zhu, G. Li, J. Hu, Nucleophilic fluoromethylation of aldehydes with fluorobis(phenylsulfonyl)methane: the importance of

- strong Li–O coordination and fluorine substitution for C–C bond formation, *Angew. Chem. Int. Ed.* 50 (2011) 2588–2592, <https://doi.org/10.1002/anie.201006931>.
- [25] C. Ni, Y.-a. Li, J. Hu, Nucleophilic fluoroalkylation of epoxides with fluorinated sulfones, *J. Org. Chem.* 71 (18) (2006) 6829–6833, <https://doi.org/10.1021/jo060955l>.
 - [26] L.M. Pratt, B. Ramachandran, J.D. Xidos, C.J. Cramer, D.G. Truhlar, Structures and aggregation states of fluoromethyl lithium and chloromethyl lithium carbenoids in the gas phase and in ethereal solvent, *J. Org. Chem.* 67 (22) (2002) 7607–7612, <https://doi.org/10.1021/jo026022g>.
 - [27] D.C. Kail, P. Malova Krizkova, A. Wiczorek, F. Hammerschmidt, On the configurational stability of chiral, nonracemic fluoro- and iodo-[D1] methylolithiums, *Chem. Eur. J.* 20 (2014) 4086–4091, <https://doi.org/10.1002/chem.201304766>.
 - [28] G. Parisi, M. Colella, S. Monticelli, G. Romanazzi, W. Holzer, T. Langer, L. Degennaro, V. Pace, R. Luisi, Exploiting a “Beast” in carbenoid chemistry: development of a straightforward direct nucleophilic fluoromethylation strategy, *J. Am. Chem. Soc.* 139 (39) (2017) 13648–13651, <https://doi.org/10.1021/jacs.7b07891>.
 - [29] F.L.M. Pattison, R.L. Buchanan, F.H. Dean, The synthesis of α -monofluoroalkanoic acids, *Can. J. Chem.* 43 (1965) 1700–1713, <https://doi.org/10.1139/v65-224>.
 - [30] F.J. Leeper, M. Rock, The synthesis of a fluorinated analogue of 5-aminolaevulinic acid, a potential inhibitor of porphyrin biosynthesis, *J. Fluorine Chem.* 51 (3) (1991) 381–396.
 - [31] F. Zhu, P.-W. Xu, F. Zhou, C.-H. Wang, J. Zhou, Recycle waste salt as reagent: a one-pot substitution/Krapcho reaction sequence to α -fluorinated esters and sulfones, *Org. Lett.* 17 (2015) 972–975, <https://doi.org/10.1021/acs.orglett.5b00072>.
 - [32] M. Ihara, N. Taniguchi, T. Kai, K. Satoh, K. Fukumoto, Enantioselective construction of quaternary stereogenic centres possessing a fluorine atom, *J. Chem. Soc., Perkin Trans. 1* (1992) 221–227, <https://doi.org/10.1039/p19920000221>.
 - [33] S. Hong, M. Kim, M. Jung, M.W. Ha, M. Lee, Y. Park, M.-H. Kim, T.-S. Kim, J. Lee, H.-G. Park, Enantioselective synthesis of α -halo- α -alkylmalonates via phase-transfer catalytic α -alkylation, *Org. Biomol. Chem.* 12 (2014) 1510–1517, <https://doi.org/10.1039/c3ob42107d>.
 - [34] J. Hutchinson, G. Sandford, J.F.S. Vaughan, Alkylation and decarboxylation of ethyl 2-fluoro-3-oxobutanoate as a route to functionalised α -fluoro-ketones, *Tetrahedron* 54 (1998) 2867–2876, [https://doi.org/10.1016/s0040-4020\(98\)83023-8](https://doi.org/10.1016/s0040-4020(98)83023-8).
 - [35] C. Ding, K. Maruoka, Enantioselective alkylation of α -fluoro- β -keto esters by asymmetric phase-transfer catalysis, *Synlett* 2009 (2009) 664–666, <https://doi.org/10.1055/s-0028-1087813>.
 - [36] C. Molina, A. Ortega-Martínez, J.M. Sansano, C. Nájera, Synthesis of 3-substituted 3-fluoro-2-oxindoles by decalytic alkylation, *Org. Biomol. Chem.* 17 (3) (2019) 482–489.
 - [37] Z.-J. Wu, H.-C. Xu, Synthesis of C3-fluorinated oxindoles through reagent-free cross-dehydrogenative coupling, *Angew. Chem. Int. Ed.* 56 (17) (2017) 4734–4738, <https://doi.org/10.1002/anie.201701329>.
 - [38] J. Villieras, D. Payan, Y. Anguelova, J.-F. Normant, Enolates sodiques α halogènes. Cas des esters et des amides, *J. Organomet. Chem.* 42 (1972) C5–C8, [https://doi.org/10.1016/s0022-328x\(00\)81802-1](https://doi.org/10.1016/s0022-328x(00)81802-1).
 - [39] T. Fuchigami, T. Hayashi, K. Akinori, Electrolytic partial fluorination of organic compounds. 3. Regioselective anodic monofluorination of organoselenium compounds and their synthetic application, *Tetrahedron Lett.* 33 (22) (1992) 3161–3164.
 - [40] B.V. Varun, K.R. Prabhu, Sulfur assisted tandem electrophilic fluorinative decylation: synthesis of α -fluoro β -ketosulfides, *J. Org. Chem.* 82 (2017) 9525–9536, <https://doi.org/10.1021/acs.joc.7b01547>.
 - [41] A.G. Myers, L. McKinstry, J.L. Gleason, Synthesis of tertiary alkyl fluoride centers by asymmetric C–C(F) bond formation, *Tetrahedron Lett.* 38 (40) (1997) 7037–7040.
 - [42] A.G. Myers, L. McKinstry, J.K. Barbay, J.L. Gleason, Practical methodology for the asymmetric synthesis of organofluorine compounds, *Tetrahedron Lett.* 39 (11) (1998) 1335–1338.
 - [43] C. Calata, J.-M. Catel, E. Pfund, T. Lequeux, Scope and limitations of the Julia-Kocienski reaction with fluorinated sulfonyl esters, *Tetrahedron* 65 (20) (2009) 3967–3973, <https://doi.org/10.1016/j.tet.2009.03.041>.
 - [44] F. Larnaud, E. Pfund, B. Linclau, T. Lequeux, Decarboxylation of fluorosulfones for the preparation fluoroalkylidene precursors, *J. Fluorine Chem.* 134 (2012) 128–135, <https://doi.org/10.1016/j.jfluchem.2011.03.005>.
 - [45] T. My Thu Truong-Nguyen, H. Togo, M. Schlosser, A fluoroisoprenylation sequence employing 2-fluoroalkenal derived azomethines as key intermediates: a stereocontrolled synthesis of 2-fluorogeraniol and 2-fluorofarnesol, *Tetrahedron* 50 (26) (1994) 7827–7836.
 - [46] T.B. Patrick, S. Hosseini, S. Bains, 2-Fluoromethyl-4,4,6-trimethyl-1,3-oxasine as a new reagent for the preparation of α -fluoroaldehydes, *Tetrahedron Lett.* 31 (1990) 179–182, [https://doi.org/10.1016/s0040-4039\(00\)94364-8](https://doi.org/10.1016/s0040-4039(00)94364-8).
 - [47] J.T. Welch, K.W. Seper, Synthesis, regioselective deprotonation, and stereoselective alkylation of fluoro ketimines, *J. Org. Chem.* 53 (13) (1988) 2991–2999, <https://doi.org/10.1021/jo00248a017>.
 - [48] A. Tarui, N. Kawashima, T. Kawakita, K. Sato, M. Omote, A. Ando, Direct and diastereoselective alkylation and aldol reactions of α -bromo- α -fluoro- β -lactams, *J. Org. Chem.* 78 (2013) 7903–7911, <https://doi.org/10.1021/jo401072h>.
 - [49] J.T. Welch, K. Araki, R. Kaweck, J.A. Wichtowski, Stereocontrol by the carbon-fluorine bond and its application to asymmetric synthesis of 3-fluoro β -lactams, *J. Org. Chem.* 58 (1993) 2454–2462, <https://doi.org/10.1021/jo00061a019>.
 - [50] A. Thenappan, D.J. Burton, A facile preparation of ethyl α -fluoroalkanoates, *Tetrahedron Lett.* 30 (1989) 3641–3644, [https://doi.org/10.1016/s0040-4039\(01\)80469-x](https://doi.org/10.1016/s0040-4039(01)80469-x).
 - [51] A. Thenappan, D.J. Burton, Alkylation of (fluorocarboethoxymethylene)tri-*n*-butylphosphorane: a facile entry to α -fluoroalkanoates, *J. Org. Chem.* 55 (1990) 2311–2317, <https://doi.org/10.1021/jo00295a016>.
 - [52] P. Beier, S. Opekar, M. Zibinsky, I. Bychinskaya, G.K.S. Prakash, A new route to α -alkyl- α -fluoromethylenebisphosphonates, *Org. Biomol. Chem.* 9 (2011) 4035–4038, <https://doi.org/10.1039/c1ob05095h>.
 - [53] G.M. Blackburn, M.J. Parratt, The synthesis of α -fluoroalkylphosphonates, *J. Chem. Soc., Perkin Trans. 1* (1986) 1425–1430, <https://doi.org/10.1039/p19860001425>.
 - [54] X. Zhang, W. Qiu, D.J. Burton, The preparation of (EtO)2P(O)CFH2ZnBr and (EtO)2P(O)CFH2Cu and their utility in the preparation of functionalized α -fluorophosphonates, *Tetrahedron Lett.* 40 (1999) 2681–2684, [https://doi.org/10.1016/s0040-4039\(99\)00305-6](https://doi.org/10.1016/s0040-4039(99)00305-6).
 - [55] R. Waschbüsch, J. Carran, P. Savignac, A new route to α -fluoroalkylphosphonates, *J. Chem. Soc., Perkin Trans. 1* (1997) 1135–1140, <https://doi.org/10.1039/a607288g>.
 - [56] C. Patois, P. Savignac, A new route to α -fluoromethyl- and α -fluoroalkylphosphonates, *J. Chem. Soc., Chem. Commun.* (1993) 1711–1712, <https://doi.org/10.1039/c39930001711>.
 - [57] T. Koizumi, T. Hagi, Y. Horie, Y. Takeuchi, Diethyl 1-fluoro-1-phenylsulfonylethylmethane phosphonate, a versatile agent for the preparation of monofluorinated building blocks, *Chem. Pharm. Bull.* 35 (9) (1987) 3959–3962, <https://doi.org/10.1248/cpb.35.3959>.
 - [58] S. Opekar, R. Pohl, P. Beran, L. Rulišek, P. Beier, Diethyl fluoronitromethylphosphonate: synthesis and application in nucleophilic fluoroalkyl additions, *Chem. Eur. J.* 20 (5) (2014) 1453–1458, <https://doi.org/10.1002/chem.201303817>.
 - [59] H.-J. Tsai, Application of fluorocarboethoxy-substituted phosphonate: a facile entry to substituted 2-fluoro-3-oxoesters, *Phosphorus Sulfur Silicon Relat. Elem.* 126 (1) (1997) 1–10, <https://doi.org/10.1080/10426509708043541>.
 - [60] H.-J. Tsai, C.-W. Hsieh, Tetrafluorine-containing ketones and acetoacetates: synthesis and mechanistic study, *J. Chin. Chem. Soc.* 54 (3) (2007) 749–757, <https://doi.org/10.1002/jccs.200700107>.
 - [61] Y. Yokoyama, Germyl anion species-promoted formation of cyanofluoromethylene compounds: first and efficient synthesis of fluorinated homoallylic and homoprop-2-ynyl cyanides, *Chem. Commun.* (1998) 1093–1094, <https://doi.org/10.1039/a802005a>.
 - [62] Y. Yokoyama, S. Suzuki, H. Furihata, S. Takahi, M. Nomura, M. Kajitani, The development of a novel synthetic method for cyanofluoroamides using a cobaltadithiolene complex, *Synthesis* 2004 (05) (2004) 701–705, <https://doi.org/10.1055/s-2004-815960>.
 - [63] P. Bravo, E. Piovosi, G. Resnati, G. Fronza, Asymmetric synthesis and structural analysis of 5-O-benzoyl-2,3-dideoxy-3-fluoro- α,β -D-ribofuranose and -xylofuranose from homochiral 1-fluoro-3-sulfinylacetone, *J. Org. Chem.* 54 (1989) 5171–5176, <https://doi.org/10.1021/jo00282a039>.
 - [64] P. Bravo, G. Resnati, F. Viani, A. Arnone, Homochiral 2,5-disubstituted 3-fluorotetrahydrofurans carrying differently functionalized ring appendages, *J. Chem. Soc., Perkin Trans. 1* (1989) 839–840, <https://doi.org/10.1039/p19890000839>.
 - [65] Y. Arroyo, M.A. Sanz-Tejedor, A. Parra, I. Alonso, J.L. Garcia Ruano, Asymmetric nucleophilic monofluorobenzoylation of allyl and propargyl halides mediated by a remote sulfinyl group: synthesis of homoallylic and homopropargylic fluorides, *J. Org. Chem.* 79 (2014) 6970–6977, <https://doi.org/10.1021/jo501096f>.
 - [66] B. Jiang, Z.-G. Huang, K.-J. Cheng, Preparation of N-phenyl-(S)-prolinol-derived P, N-ligands and their application in Pd-catalyzed asymmetric allylic alkylation, *Tetrahedron: Asymmetry* 17 (6) (2006) 942–951, <https://doi.org/10.1016/j.tetasy.2006.03.010>.
 - [67] K. Shibamoto, T. Muto, Y. Sumikawa, A. Narayama, S. Iwasa, Development of a new chiral spiro oxazolinyldipyrroline ligand (Spymox) for asymmetric catalysis, *Synlett* 2009 (02) (2009) 241–244, <https://doi.org/10.1055/s-0028-1087675>.
 - [68] N. Gao, X.-M. Zhao, C.-S. Cai, J.-W. Cai, Enantioselective synthesis of monofluorinated allylic compounds: Pd-catalyzed asymmetric allylations of dimethyl 2-fluoromalonate using new N-sulfinyl-based ligands, *Org. Biomol. Chem.* 13 (37) (2015) 9551–9558.
 - [69] T. Fukuzumi, N. Shibata, M. Sugiura, H. Yasui, S. Nakamura, T. Toru, Fluorobis (phenylsulfonylethyl)methane: a fluoromethide equivalent and palladium-catalyzed enantioselective allylic monofluoromethylation, *Angew. Chem. Int. Ed.* 45 (30) (2006) 4973–4977, <https://doi.org/10.1002/anie.200600625>.
 - [70] Y. Huang, Q.-S. Zhang, P. Fang, T.-G. Chen, J. Zhu, X.-L. Hou, Pd-catalyzed highly regio-, diastereo-, and enantioselective allylic alkylation of α -fluorophosphonates, *Chem. Commun.* 50 (2014) 6751–6753, <https://doi.org/10.1039/c4cc02158d>.
 - [71] M. Ogasawara, H. Murakami, T. Furukawa, T. Takahashi, N. Shibata, Synthesis of fluorinated allenes via palladium-catalyzed monofluoromethylation using FBSM, *Chem. Commun.* (2009) 7366–7368, <https://doi.org/10.1039/b912531k>.

- [72] W.-B. Liu, S.-C. Zheng, H. He, X.-M. Zhao, L.-X. Dai, S.-L. You, Iridium-catalyzed regio- and enantioselective allylic alkylation of fluorobis(phenylsulfonyl) methane, *Chem. Commun.* (2009) 6604–6606, <https://doi.org/10.1039/b914315g>.
- [73] H. Zhang, J. Chen, X.-M. Zhao, Enantioselective synthesis of fluorinated branched allylic compounds via Ir-catalyzed allylations of functionalized fluorinated methylene derivatives, *Org. Biomol. Chem.* 14 (30) (2016) 7183–7186.
- [74] D.J. Burton, L. Lu, Fluorinated organometallic compounds, in: R.D. Chambers (Ed.), *Topics in Current Chemistry*, Springer, Berlin Heidelberg, Berlin, Heidelberg, 1997, pp. 45–89.
- [75] M. Obayashi, E. Ito, K. Matsui, K. Kondo, (Diethylphosphinyl) difluoromethyl lithium. – preparation and synthetic application, *Tetrahedron Lett.* 23 (1982) 2323–2326, [https://doi.org/10.1016/S0040-4039\(00\)87332-3](https://doi.org/10.1016/S0040-4039(00)87332-3).
- [76] G.K.S. Prakash, C. Ni, F. Wang, J. Hu, G.A. Olah, From difluoromethyl 2-pyridyl sulfone to difluorinated sulfonates: a protocol for nucleophilic difluoro (sulfonato)methylation, *Angew. Chem. Int. Ed.* 50 (11) (2011) 2559–2563, <https://doi.org/10.1002/anie.201007594>.
- [77] L. Zhu, Y.a. Li, Y. Zhao, J. Hu, Nucleophilic (phenylsulfonyl) difluoromethylation of alkyl halides using PhSO₂CF₂SiMe₃: preparation of gem-difluoroalkenes and trifluoromethyl compounds, *Tetrahedron Lett.* 51 (47) (2010) 6150–6152, <https://doi.org/10.1016/j.tetlet.2010.09.068>.
- [78] C. Portella, T. Brigaud, O. Lefebvre, R. Plantier-Royon, Convergent synthesis of fluoro and gem-difluoro compounds using trifluoromethyltrimethylsilane, *J. Fluorine Chem.* 101 (2) (2000) 193–198.
- [79] K.T. O'Brien, J.W. Nadraws, A.B. Smith, A difluoromethylene linchpin/synthon: application in conjunction with anion relay chemistry (ARC) permits ready access to diverse difluoromethylene scaffolds, *Org. Lett.* 23 (5) (2021) 1521–1524, <https://doi.org/10.1021/acs.orglett.0c03508>.
- [80] B.W. Metcalf, E.T. Jarvi, J.P. Burkhart, The synthesis of α,α -difluoroaldehydes and ketones via claisen rearrangements, *Tetrahedron Lett.* 26 (1985) 2861–2864, [https://doi.org/10.1016/S0040-4039\(00\)98856-7](https://doi.org/10.1016/S0040-4039(00)98856-7).
- [81] C. Audouard, M.R. Garayt, E. Kerouedan, J.M. Percy, H. Yang, A direct and rapid route to α,α -difluoroacetyl silanes from trifluoroethanol, *J. Fluorine Chem.* 126 (2005) 609–621, <https://doi.org/10.1016/j.jfluchem.2005.01.005>.
- [82] V. Reutrakul, T. Kruahong, M. Pohmakotr, A novel method for the synthesis of α -fluoroketones via Claisen rearrangement, *Tetrahedron Lett.* 35 (1994) 4853–4856, [https://doi.org/10.1016/S0040-4039\(00\)76986-3](https://doi.org/10.1016/S0040-4039(00)76986-3).
- [83] D.J. Burton, R. Takei, S. Shin-Ya, Preparation, stability, reactivity and synthetic utility of a cadmium stabilized complex of difluoromethylene phosphonic acid ester, *J. Fluorine Chem.* 18 (2) (1981) 197–202.
- [84] R.D. Chambers, R. Jaouhari, D. O'Hagan, Fluorine in enzyme chemistry: part 1. Synthesis of difluoromethylenephosphonate derivatives as phosphate mimics, *J. Fluorine Chem.* 44 (2) (1989) 275–284.
- [85] R.D. Chambers, R. Jaouhari, D. O'Hagan, Synthesis of a difluoromethylenephosphonate analogue of glycerol-3-phosphate. A substrate for NADH linked glycerol-3-phosphate dehydrogenase, *J. Chem. Soc., Chem. Commun.* (1988) 1169–1170, <https://doi.org/10.1039/c39880001169>.
- [86] D.J. Burton, D.M. Wiemers, A remarkably simple preparation of (trifluoromethyl)cadmium and -zinc reagents directly from difluorodihalomethanes, *J. Am. Chem. Soc.* 107 (17) (1985) 5014–5015, <https://doi.org/10.1021/ja00303a042>.
- [87] G.A. Hartgraves, D.J. Burton, The preparation and allylation of difluoromethylcadmium, *J. Fluorine Chem.* 39 (3) (1988) 425–430.
- [88] D.J. Burton, G.A. Hartgraves, The preparation of HCF₂CdX and HCF₂ZnX via direct insertion into the carbon halogen bond of CF₂HY (Y=Br, I), *J. Fluorine Chem.* 128 (10) (2007) 1198–1215, <https://doi.org/10.1016/j.jfluchem.2007.05.015>.
- [89] C.A. Falcioia, A. Alexakis, Copper-catalyzed asymmetric allylic alkylation, *Eur. J. Org. Chem.* 2008 (22) (2008) 3765–3780, <https://doi.org/10.1002/ejoc.200800025>.
- [90] T. Kawasaki, T. Kitazume, Reaction of organometallic reagents of ethyl bromodifluoroacetate with allylic halides, *J. Fluorine Chem.* 88 (1) (1998) 99–103.
- [91] T. Taguchi, O. Kitagawa, T. Morikawa, T. Nishiwaki, H. Uehara, H. Endo, Y. Kobayashi, Synthesis of 2,2-difluoroesters by iododifluoroacetate-copper with organic halides, *Tetrahedron Lett.* 27 (50) (1986) 6103–6106.
- [92] O. Kitagawa, T. Taguchi, Y. Kobayashi, (Methyl difluoroacetate)copper reagent. Remarkable solvent effect on the 19F-NMR spectrum, stability and reactivity, *Chem. Lett.* 18 (3) (1989) 389–392, <https://doi.org/10.1246/cl.1989.389>.
- [93] F. Chevrier, Z. Chamas, T. Lequeux, E. Pfund, G. Andrei, R. Snoeck, V. Roy, L.A. Agrofoglio, Synthesis of 5,5-difluoro-5-phosphono-pent-2-en-1-yl nucleosides as potential antiviral agents, *RSC Adv.* 7 (51) (2017) 32282–32287.
- [94] D. Barte, M.J. Wheadon, C.L. Freely Meyers, Synthesis and evaluation of fluoroalkyl phosphonyl analogues of 2-C-methylerythritol phosphate as substrates and inhibitors of IspD from human pathogens, *J. Org. Chem.* 83 (17) (2018) 9580–9591, <https://doi.org/10.1021/acs.joc.8b00686>.
- [95] S. Hikishima, M. Hashimoto, L. Magnowska, A. Bzowska, T. Yokomatsu, Structural-based design and synthesis of novel 9-deazaguanine derivatives having a phosphate mimic as multi-substrate analogue inhibitors for mammalian PNPs, *Bioorg. Med. Chem.* 18 (6) (2010) 2275–2284, <https://doi.org/10.1016/j.bmc.2010.01.062>.
- [96] A.M. Kawamoto, M.M. Campbell, Novel class of difluorovinylphosphonate analogues of PEP, *J. Chem. Soc., Perkin Trans. 1* (1997) 1249–1254, <https://doi.org/10.1039/ja603094g>.
- [97] A.D. Tiwari, Y. Guan, D.R. Grabowski, J.P. Maciejewski, B.K. Jha, J.G. Phillips, SAR insights into TET2 catalytic domain inhibition: synthesis of 2-Hydroxy-4-Methylene-pentenedicarboxylates, *Bioorg. Med. Chem.* 39 (2021) 116141, <https://doi.org/10.1016/j.bmc.2021.116141>.
- [98] T. Yokomatsu, A. Ichimura, J. Kato, S. Shibuya, Synthesis of allenic (α,α -difluoromethylene)phosphonates from propargylic tosylates and acetates, *Synlett* 2001 (2001) 287–289, <https://doi.org/10.1055/s-2001-10795>.
- [99] S. Hikishima, M. Hashimoto, L. Magnowska, A. Bzowska, T. Yokomatsu, Synthesis and biological evaluation of 9-deazaguanine derivatives connected by a linker to difluoromethylene phosphonic acid as multi-substrate analogue inhibitors of PNP, *Bioorg. Med. Chem. Lett.* 17 (15) (2007) 4173–4177, <https://doi.org/10.1016/j.bmcl.2007.05.054>.
- [100] M.V. Ivanova, A. Bayle, T. Besset, X. Pannecoucke, T. Poisson, Copper-mediated introduction of the CF₂PO(OEt)₂ motif: scope and limitations, *Chem. Eur. J.* 23 (2017) 17318–17338, <https://doi.org/10.1002/chem.201703542>.
- [101] D.J. Burton, L.G. Sprague, Allylations of [(diethoxyphosphinyl)difluoromethyl] zinc bromide as a convenient route to 1,1-difluoro-3-alkenephosphonates, *J. Org. Chem.* 54 (3) (1989) 613–617, <https://doi.org/10.1021/jo00264a021>.
- [102] X. Liu, H. Chen, E. Laurini, Y. Wang, V. Dal Col, P. Posocco, F. Ziarelli, M. Fermeglia, C.-C. Zhang, S. Pricl, L. Peng, 2-Difluoromethylene-4-methylenepentanoic acid, a paradoxical probe able to mimic the signaling role of 2-oxoglutaric acid in cyanobacteria, *Org. Lett.* 13 (11) (2011) 2924–2927, <https://doi.org/10.1021/ol2009544>.
- [103] B.P. Martin, E. Vasilieva, C.M. Dupureur, C.D. Spilling, Synthesis and comparison of the biological activity of monocyclic phosphonate, difluorophosphonate and phosphate analogs of the natural AChE inhibitor cyclophostin, *Bioorg. Med. Chem.* 23 (24) (2015) 7529–7534, <https://doi.org/10.1016/j.bmc.2015.10.044>.
- [104] K. Harada, J. Mizukami, T. Watanabe, G. Mori, M. Ubukata, K. Suwa, S. Fukuda, T. Negoro, M. Sato, T. Inaba, Optimization of oxadiazole derivatives with a spirocyclic cyclohexane structure as novel GPR119 agonists, *Bioorg. Med. Chem. Lett.* 29 (16) (2019) 2100–2106, <https://doi.org/10.1016/j.bmcl.2019.07.004>.
- [105] H. Kadri, T.E. Taher, Q. Xu, M. Sharif, E. Ashby, R.T. Bryan, B.E. Willcox, Y. Mehellou, Aryloxy diester phosphonamide prodrugs of phosphoantigens (ProPAGens) as potent activators of VO₂/VO₂2 T-cell immune responses, *J. Med. Chem.* 63 (2020) 11258–11270, <https://doi.org/10.1021/acs.jmedchem.0c01232>.
- [106] L. Xu, D.A. Vici, Direct difluoromethylation of aryl halides via base metal catalysis at room temperature, *J. Am. Chem. Soc.* 138 (8) (2016) 2536–2539, <https://doi.org/10.1021/jacs.6b00053>.
- [107] K. Aikawa, K. Ishii, Y. Endo, K. Mikami, Copper-catalyzed allylic difluoromethylation of allyl carbonates with (difluoromethyl)zinc reagent, *J. Fluorine Chem.* 203 (2017) 122–129, <https://doi.org/10.1016/j.jfluchem.2017.07.018>.
- [108] V.V. Levin, A.A. Zemtsov, M.I. Struchkova, A.D. Dilman, Reactions of difluorocarbene with organozinc reagents, *Org. Lett.* 15 (4) (2013) 917–919, <https://doi.org/10.1021/ol400122k>.
- [109] V.V. Levin, A.A. Zemtsov, M.I. Struchkova, A.D. Dilman, Reactions of organozinc reagents with potassium bromodifluoroacetate, *J. Fluorine Chem.* 171 (2015) 97–101, <https://doi.org/10.1016/j.jfluchem.2014.08.021>.
- [110] A.A. Zemtsov, N.S. Kondratyev, V.V. Levin, M.I. Struchkova, A.D. Dilman, Copper-catalyzed allylation of α,α -difluoro-substituted organozinc reagents, *J. Org. Chem.* 79 (2014) 818–822, <https://doi.org/10.1021/jo4024705>.
- [111] A.A. Zemtsov, N.S. Kondratyev, V.V. Levin, M.I. Struchkova, A.D. Dilman, Reactions of gem-difluoro-substituted organozinc reagents with propargyl halides, *Russ. Chem. Bull.* 65 (11) (2016) 2760–2762, <https://doi.org/10.1007/s11172-016-1649-8>.
- [112] M.D. Kosobokov, V.V. Levin, A.A. Zemtsov, M.I. Struchkova, A.A. Korlyukov, D. E. Arkhipov, A.D. Dilman, Geminal silicon/zinc reagent as an equivalent of difluoromethylene bis-carbanion, *Org. Lett.* 16 (5) (2014) 1438–1441, <https://doi.org/10.1021/ol5002297>.
- [113] V.V. Levin, V.O. Smirnov, M.I. Struchkova, A.D. Dilman, Nucleophilic Iododifluoromethylation of Aldehydes Using Bromine/Iodine Exchange, *J. Org. Chem.* 80 (18) (2015) 9349–9353, <https://doi.org/10.1021/acs.joc.5b01590>.
- [114] K. Uneyama, G.O. Mizutani, K. Maeda, T. Kato, Electroreductive defluorination of trifluoromethyl ketones and trifluoroacetic acid derivatives, *J. Org. Chem.* 64 (18) (1999) 6717–6723, <https://doi.org/10.1021/jo990571d>.
- [115] G.B. Hammond, Nucleophilic and electrophilic substitutions of difluoropropargyl bromides, *J. Fluorine Chem.* 127 (4–5) (2006) 476–488, <https://doi.org/10.1016/j.jfluchem.2005.12.024>.
- [116] M. Mae, J.A. Hong, G.B. Hammond, K. Uneyama, Mg(0)-promoted debromometalation of gem-difluoropropargyl bromides, *Tetrahedron Lett.* 46 (11) (2005) 1787–1789, <https://doi.org/10.1016/j.tetlet.2005.01.130>.
- [117] M.-H. Yang, D.L. Orsi, R.A. Altman, Ligand-controlled regioselective palladium-catalyzed decarboxylative allylation reaction to access α,α -difluoroketones, *Angew. Chem. Int. Ed.* 54 (8) (2015) 2361–2365, <https://doi.org/10.1002/anie.201410039>.
- [118] D.J. Burton, Z.-Y. Yang, Fluorinated organometallics: Perfluoroalkyl and functionalized perfluoroalkyl organometallic reagents in organic synthesis,

- Tetrahedron 48 (2) (1992) 189–275, [https://doi.org/10.1016/S0040-4020\(01\)88139-4](https://doi.org/10.1016/S0040-4020(01)88139-4).
- [119] G.K.S. Prakash, F. Wang, Z. Zhang, R. Haiges, M. Rahm, K.O. Christe, T. Mathew, G.A. Olah, Long-lived trifluoromethanide anion: a key intermediate in nucleophilic trifluoromethylations, *Angew. Chem. Int. Ed.* 53 (43) (2014) 11575–11578, <https://doi.org/10.1002/anie.201406505>.
- [120] A. Haas, J. Koehler, Darstellung von polyfluororganotrichlorsilanen, *J. Fluorine Chem.* 17 (6) (1981) 531–537.
- [121] D.J. Burton, Y. Tarumi, P.L. Heinze, The preparation and allylation of perfluoroalkyl cadmium and copper reagents, *J. Fluorine Chem.* 50 (2) (1990) 257–263.
- [122] D.J. Burton, Z.-Y. Yang, V. Platonov, The preparation, stability and reactivity of perfluorobenzyl-cadmium and -copper reagents, *J. Fluorine Chem.* 66 (1) (1994) 23–24, [https://doi.org/10.1016/0022-1139\(93\)02919-6](https://doi.org/10.1016/0022-1139(93)02919-6).
- [123] G. Bubot, D. Mansuy, S. Lecolier, J.F. Normant, Action du perfluoroisopropylargent sur les halogénures d'alcyle et l'anhydride carbonique, *J. Organomet. Chem.* 42 (1972) C105–C106, [https://doi.org/10.1016/S0022-328X\(00\)90069-X](https://doi.org/10.1016/S0022-328X(00)90069-X).
- [124] C. Chuit, R.J.P. Corriu, C. Rye, J.C. Young, Reactivity of penta- and hexacoordinate silicon compounds and their role as reaction intermediates, *Chem. Rev.* 93 (4) (1993) 1371–1448, <https://doi.org/10.1021/cr00020a003>.
- [125] A.D. Dilman, V.V. Levin, M. Karni, Y. Apeloig, Activation of pentafluorophenylsilanes by weak Lewis bases in reaction with iminium cations, *J. Org. Chem.* 71 (19) (2006) 7214–7223, <https://doi.org/10.1021/jo0606812>.
- [126] A.D. Dilman, V.V. Levin, A.A. Korlyukov, P.A. Belyakov, M.I. Struchkova, M.Y. Antipin, V.A. Tartakovsky, Complexation of tris(pentafluorophenyl)silanes with neutral Lewis bases, *J. Organomet. Chem.* 693 (6) (2008) 1005–1019, <https://doi.org/10.1016/j.jorganchem.2007.12.022>.
- [127] A.A. Zemtsov, V.V. Levin, A.D. Dilman, M.I. Struchkova, P.A. Belyakov, V.A. Tartakovsky, J. Hu, Reaction of Baylis-Hillman adducts with fluorinated silanes, *Eur. J. Org. Chem.* 2010 (35) (2010) 6779–6785, <https://doi.org/10.1002/ejoc.201001051>.
- [128] T. Furukawa, T. Nishimine, E. Tokunaga, K. Hasegawa, M. Shiro, N. Shibata, Organocatalyzed regio- and enantioselective allylic trifluoromethylation of Morita-Baylis-Hillman adducts using Ruppert-Prakash reagent, *Org. Lett.* 13 (15) (2011) 3972–3975, <https://doi.org/10.1021/ol201490e>.
- [129] Y. Li, F. Liang, Q. Li, Y.-C. Xu, Q.-R. Wang, L. Jiang, Room temperature asymmetric allylic trifluoromethylation of Morita-Baylis-Hillman carbonates, *Org. Lett.* 13 (22) (2011) 6082–6085, <https://doi.org/10.1021/ol202572u>.
- [130] T.-Y. Liu, M. Xie, Y.-C. Chen, Organocatalytic asymmetric transformations of modified Morita-Baylis-Hillman adducts, *Chem. Soc. Rev.* 41 (2012) 4101–4112, <https://doi.org/10.1039/C2CS35017C>.
- [131] T. Nishimine, K. Fukushi, N. Shibata, H. Taira, E. Tokunaga, A. Yamano, M. Shiro, N. Shibata, Kinetic resolution of allyl fluorides by enantioselective allylic trifluoromethylation based on silicon-assisted C-F bond cleavage, *Angew. Chem. Int. Ed.* 53 (2014) 517–520, <https://doi.org/10.1002/anie.201308071>.
- [132] T. Kitazume, N. Ishikawa, Palladium-catalyzed cross-coupling reactions between allyl, vinyl or aryl halide and perfluoroalkyl iodide with zinc and ultrasonic irradiation, *Chem. Lett.* 11 (1) (1982) 137–140, <https://doi.org/10.1246/cl.1982.137>.
- [133] T. Kitazume, N. Ishikawa, Ultrasound-promoted selective perfluoroalkylation on the desired position of organic molecules, *J. Am. Chem. Soc.* 107 (18) (1985) 5186–5191, <https://doi.org/10.1021/ja00304a026>.
- [134] B.M. Trost, H. Gholami, D. Zell, Palladium-catalyzed asymmetric allylic fluoroalkylation/trifluoromethylation, *J. Am. Chem. Soc.* 141 (29) (2019) 11446–11451, <https://doi.org/10.1021/jacs.9b06231>.
- [135] G.G. Dubinina, H. Furutachi, D.A. Vicić, Active trifluoromethylating agents from well-defined copper(I)–CF₃ complexes, *J. Am. Chem. Soc.* 130 (27) (2008) 8600–8601, <https://doi.org/10.1021/ja802946s>.
- [136] H. Morimoto, T. Tsubogo, N.D. Litvinas, J.F. Hartwig, A broadly applicable copper reagent for trifluoromethylations and perfluoroalkylations of aryl iodides and bromides, *Angew. Chem. Int. Ed.* 50 (16) (2011) 3793–3798, <https://doi.org/10.1002/anie.201100633>.
- [137] A. Zanardi, M.A. Novikov, E. Martin, J. Benet-Buchholz, V.V. Grushin, Direct cupration of fluoroform, *J. Am. Chem. Soc.* 133 (51) (2011) 20901–20913, <https://doi.org/10.1021/ja2081026>.
- [138] T. Liu, Q. Shen, Progress in copper-mediated formation of trifluoromethylated arenes, *Eur. J. Org. Chem.* 2012 (34) (2012) 6679–6687, <https://doi.org/10.1002/ejoc.201200648>.
- [139] H.-e. Liu, J. Wu, Y. Jin, X. Leng, Q. Shen, Mechanistic insight into copper-mediated trifluoromethylation of aryl halides: the role of CuI, *J. Am. Chem. Soc.* 143 (35) (2021) 14367–14378, <https://doi.org/10.1021/jacs.1c07408>.
- [140] Y. Kobayashi, K. Yamamoto, I. Kumadaki, Trifluoromethylation of aliphatic halides with trifluoromethyl copper, *Tetrahedron Lett.* 20 (42) (1979) 4071–4072.
- [141] J.M. Paratian, E. Labbé, S. Sibille, J. Périchon, Electrosynthesis of (trifluoromethyl)copper complexes from bromotrifluoromethane: reactivities with various organic halides, *J. Organomet. Chem.* 489 (1–2) (1995) 137–143.
- [142] D.J. Burton, G.A. Hartgraves, J. Hsu, A facile, general route to perfluoroalkyl allenes, *Tetrahedron Lett.* 31 (26) (1990) 3699–3702.
- [143] P.L. Coe, N.E. Milner, Perfluoroalkylcopper(I) compounds, *J. Organomet. Chem.* 70 (1) (1974) 147–152.
- [144] H. Kawai, T. Furukawa, Y. Nomura, E. Tokunaga, N. Shibata, Cu-mediated chemoselective trifluoromethylation of benzyl bromides using shelf-stable electrophilic trifluoromethylating reagents, *Org. Lett.* 13 (14) (2011) 3596–3599, <https://doi.org/10.1021/ol201205t>.
- [145] Y.-L. Ji, J.-J. Kong, J.-H. Lin, J.-C. Xiao, Y.-C. Gu, Copper-mediated trifluoromethylation of propargyl acetates leading to trifluoromethylallenes, *Org. Biomol. Chem.* 12 (2014) 2903–2906, <https://doi.org/10.1039/c3ob42575d>.
- [146] T.S.N. Zhao, K.J. Szabó, Trifluoromethylation of propargylic halides and trifluoroacetates using (Ph₃P)₃Cu(CF₃) reagent, *Org. Lett.* 14 (15) (2012) 3966–3969, <https://doi.org/10.1021/ol3017287>.
- [147] Y. Miyake, S.-I. Ota, M. Shibata, K. Nakajima, Y. Nishibayashi, Copper-catalyzed nucleophilic trifluoromethylation of propargylic halides, *Chem. Commun.* 49 (2013) 7809–7811, <https://doi.org/10.1039/c3cc44434a>.
- [148] D.M. Wiemers, D.J. Burton, Pregeneration, spectroscopic detection and chemical reactivity of (trifluoromethyl)copper, an elusive and complex species, *J. Am. Chem. Soc.* 108 (4) (1986) 832–834, <https://doi.org/10.1021/ja00264a043>.
- [149] J.-P. Bouillon, C. Maliverney, R. Merenyi, H.G. Viehe, Trifluoromethylation of aliphatic halogen compounds, *J. Chem. Soc., Perkin Trans. 1* (1991) 2147–2149, <https://doi.org/10.1039/p19910002147>.
- [150] S. Lukaszewski-Rose, D.J. Burton, Synthesis and functionalization of (Z)-1,2-difluoro-1-tri-n-butylstannyl-1,4-pentadiene, *J. Fluorine Chem.* 179 (2015) 23–32, <https://doi.org/10.1016/j.jfluchem.2014.12.006>.
- [151] Q. Liu, D.J. Burton, The stereoselective synthesis of (Z)-HFCCFZnI and stereospecific preparation of (E)-1,2-difluorostyrenes from (Z)-HFCCFZnI via an unusual Pd(PPh₃)₄–Cu(I)Br co-catalysis approach or (Z)-HFCCF₃NBu₃, *J. Fluorine Chem.* 132 (2) (2011) 78–87, <https://doi.org/10.1016/j.jfluchem.2010.11.008>.
- [152] C.P. Johnston, T.H. West, R.E. Dooley, M. Reid, A.B. Jones, E.J. King, A.G. Leach, G.C. Lloyd-Jones, Anion-initiated trifluoromethylation by TMSCF₃: deconvolution of the silicate-carbanion dichotomy by stopped-flow NMR/IR, *J. Am. Chem. Soc.* 140 (35) (2018) 11112–11124, <https://doi.org/10.1021/jacs.8b06777>.
- [153] N. Maggiora, W. Tyrre, D. Naumann, N.V. Kirij, Y.L. Yagupolskii, [Me₃Si(CF₃)F]– and [Me₃Si(CF₃)₂]–: reactive intermediates in fluoride-initiated trifluoromethylation with Me₃SiCF₃ – an NMR study, *Angew. Chem. Int. Ed.* 38 (1999) 2252–2253, [https://doi.org/10.1002/\(SICI\)1521-3773\(19990802\)38:15<2252::AID-ANIE2252>3.0.CO;2-Z](https://doi.org/10.1002/(SICI)1521-3773(19990802)38:15<2252::AID-ANIE2252>3.0.CO;2-Z).
- [154] A. Kolomeitsev, G. Bissky, E. Lork, V. Movchun, E. Rusanov, P. Kirsch, G.-V. Röschenhaler, Different fluoride anion sources and (trifluoromethyl)trimethylsilane: molecular structure of tris(dimethylamino)sulfonium bis(trifluoromethyl)trimethylsilicate, the first isolated pentacoordinate silicon species with five Si–C bonds, *Chem. Commun.* (1999) 1017–1018, <https://doi.org/10.1039/a901953g>.
- [155] D. Chen, C. Ni, Y. Zhao, X. Cai, X. Li, P. Xiao, J. Hu, Bis(difluoromethyl)trimethylsilicate anion: a key intermediate in nucleophilic difluoromethylation of enolizable ketones with Me₃SiCF₂H, *Angew. Chem. Int. Ed.* 55 (41) (2016) 12632–12636, <https://doi.org/10.1002/anie.201605280>.
- [156] H. Urata, T. Fuchikami, A novel and convenient method for trifluoromethylation of organic halides using CF₃SiR₃/KF/Cu(I) system, *Tetrahedron Lett.* 32 (1) (1991) 91–94.
- [157] J. Kim, J.N.M. Shreeve, The first Cu(I)-mediated nucleophilic trifluoromethylation reactions using (trifluoromethyl)trimethylsilane in ionic liquids, *Org. Biomol. Chem.* 2 (2004) 2728–2734, <https://doi.org/10.1039/b412480b>.
- [158] X. Jiang, F.-L. Qing, Cu-Mediated trifluoromethylation of benzyl, allyl and propargyl methanesulfonates with TMSCF₃, *Beilstein J. Org. Chem.* 9 (2013) 2862–2865, <https://doi.org/10.3762/bjoc.9.322>.
- [159] Y. Miyake, S.-I. Ota, Y. Nishibayashi, Copper-catalyzed nucleophilic trifluoromethylation of allylic halides: a simple approach to allylic trifluoromethylation, *Chem. Eur. J.* 18 (42) (2012) 13255–13258, <https://doi.org/10.1002/chem.201202853>.
- [160] X. Gao, Y.-L. Xiao, X. Wan, X. Zhang, Copper-catalyzed highly stereoselective trifluoromethylation and difluoroalkylation of secondary propargyl sulfonates, *Angew. Chem. Int. Ed.* 57 (12) (2018) 3187–3191, <https://doi.org/10.1002/anie.201711463>.
- [161] Z.Y. Yang, D.M. Wiemers, D.J. Burton, Trifluoromethylcopper: a useful difluoromethylene transfer reagent: a novel double insertion of difluoromethylene into pentafluorophenylcopper, *J. Am. Chem. Soc.* 114 (11) (1992) 4402–4403, <https://doi.org/10.1021/ja00037a057>.
- [162] Z.-Y. Yang, D.J. Burton, A novel double insertion of the difluoromethylene unit from trifluoromethylcopper into the carbon–copper bond of perfluoroaryl- and perfluorovinylcopper reagents: preparation, mechanism and applications of new fluorinated copper reagents, *J. Fluorine Chem.* 102 (1–2) (2000) 89–103.
- [163] L. Li, C. Ni, Q. Xie, M. Hu, F. Wang, J. Hu, TMSCF₃ as a convenient source of CF₂=CF₂ for pentafluoroethylation, (aryloxy)tetrafluoroethylation, and tetrafluoroethylation, *Angew. Chem. Int. Ed.* 56 (33) (2017) 9971–9975, <https://doi.org/10.1002/anie.201705734>.
- [164] J. Mestre, S. Castillón, O. Boutureira, “Ligandless” pentafluoroethylation of unactivated (hetero)aryl and alkenyl halides enabled by the controlled self-condensation of TMSCF₃-derived CuCF₃, *J. Org. Chem.* 84 (23) (2019) 15087–15097, <https://doi.org/10.1021/acs.joc.9b02001>.

- [165] Q. Xie, Z. Zhu, L. Li, C. Ni, J. Hu, Controllable double CF₂-insertion into sp² C–Cu bond using TMSCF₃: a facile access to tetrafluoroethylene-bridged structures, *Chem. Sci.* 11 (1) (2020) 276–280.
- [166] S.u. De-Bao, D. Jian-Xiang, C. Qing-Yun, Methyl chlorodifluoroacetate a convenient trifluoromethylating agent, *Tetrahedron Lett.* 32 (52) (1991) 7689–7690.
- [167] Z.-Y. Long, J.-X. Duan, Y.-B. Lin, C.-Y. Guo, Q.-Y. Chen, Potassium 3-oxa- ω -fluorosulfonylperfluoropentanoate (FO₂SCF₂CF₂OCF₂CO₂K), a low-temperature trifluoromethylating agent for organic halides; its α -carbon-oxygen bond fragmentation, *J. Fluorine Chem.* 78 (1996) 177–181, [https://doi.org/10.1016/0022-1139\(96\)03438-0](https://doi.org/10.1016/0022-1139(96)03438-0).
- [168] Q.-Y. Chen, J.-X. Duan, Methyl 3-oxo- ω -fluorosulfonylperfluoropentanoate: a versatile trifluoromethylating agent for organic halides, *J. Chem. Soc., Chem. Commun.* (1993) 1389–1391, <https://doi.org/10.1039/c39930001389>.
- [169] J.-X. Duan, Q.-Y. Chen, Novel synthesis of 2,2,2-trifluoroethyl compounds from homoallylic alcohols: a copper(I) iodide-initiated trifluoromethyl-dehydroxylation process, *J. Chem. Soc., Perkin Trans. 1* (1994) 725–730, <https://doi.org/10.1039/p19940000725>.
- [170] L. Tan, C.-y. Chen, R.D. Larsen, T.R. Verhoeven, P.J. Reider, An efficient asymmetric synthesis of a potent COX-2 inhibitor L-784,512, *Tetrahedron Lett.* 39 (23) (1998) 3961–3964.
- [171] B.R. Ambler, R.A. Altman, Copper-catalyzed decarboxylative trifluoromethylation of allylic bromodifluoroacetates, *Org. Lett.* 15 (21) (2013) 5578–5581, <https://doi.org/10.1021/ol402780k>.
- [172] B. Ambler, S. Peddi, R. Altman, Copper-catalyzed decarboxylative trifluoromethylation of propargyl bromodifluoroacetates, *Synthesis* 46 (14) (2014) 1938–1946, <https://doi.org/10.1055/s-0033-1339128>.
- [173] B.R. Ambler, S. Peddi, R.A. Altman, Ligand-controlled regioselective copper-catalyzed trifluoromethylation to generate (trifluoromethyl)allenes, *Org. Lett.* 17 (10) (2015) 2506–2509, <https://doi.org/10.1021/acs.orglett.5b01027>.
- [174] F. de Azambuja, S.M. Lovrien, P. Ross, B.R. Ambler, R.A. Altman, Catalytic one-step deoxytrifluoromethylation of alcohols, *J. Org. Chem.* 84 (4) (2019) 2061–2071, <https://doi.org/10.1021/acs.joc.8b03072>.
- [175] O.A. Tomashenko, E.C. Escudero-Adán, M. Martínez Belmonte, V.V. Grushin, and easily accessible well-defined CuCF₃ aromatic trifluoromethylating agents, *Angew. Chem. Int. Ed.* 50 (33) (2011) 7655–7659, <https://doi.org/10.1002/anie.201101577>.
- [176] J.M. Larsson, S.R. Pathipati, K.J. Szabó, Regio- and stereoselective allylic trifluoromethylation and fluorination using CuCF₃ and CuF reagents, *J. Org. Chem.* 78 (14) (2013) 7330–7336, <https://doi.org/10.1021/jo4010074>.